

BFS Traversal

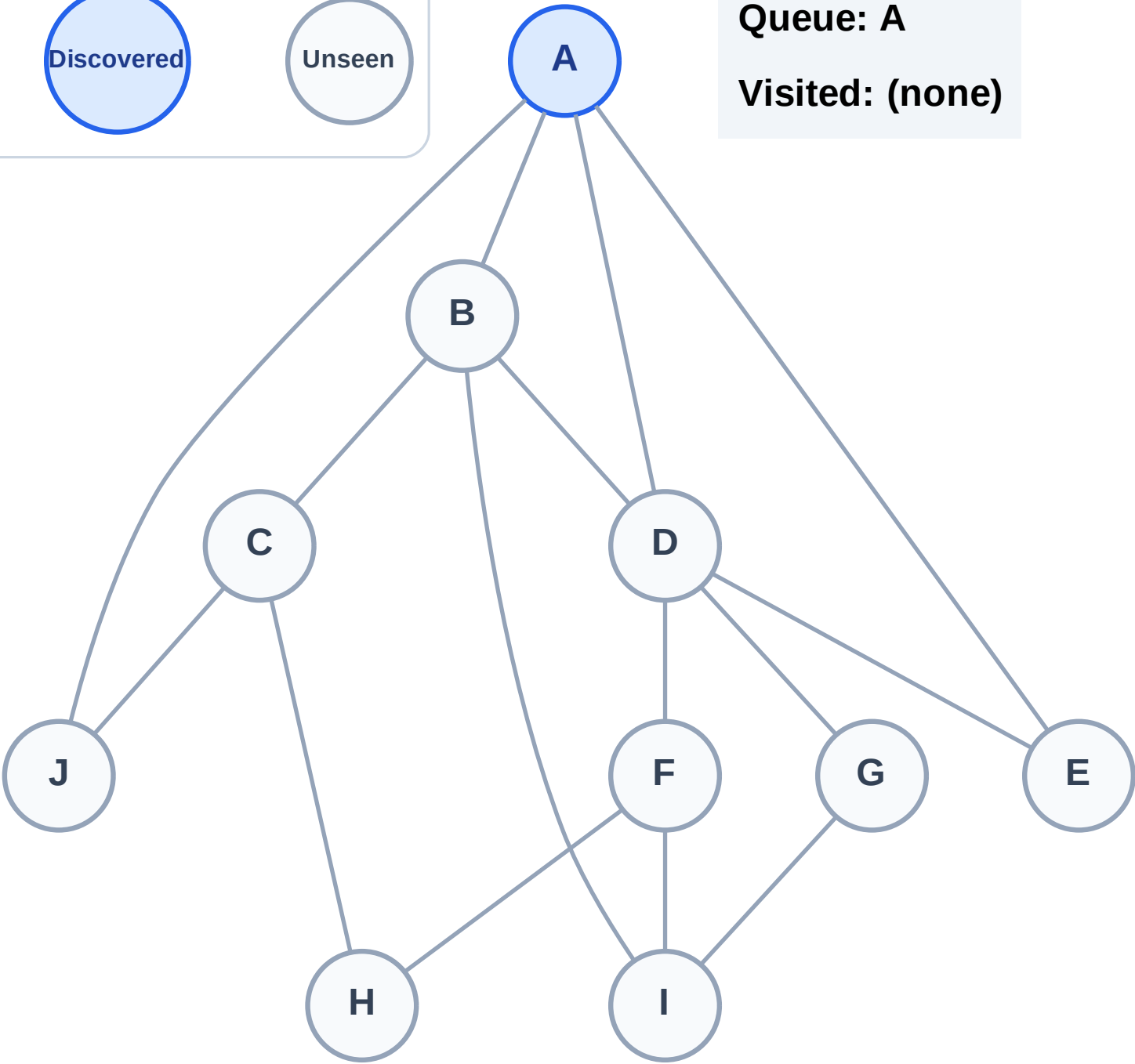
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

Complete

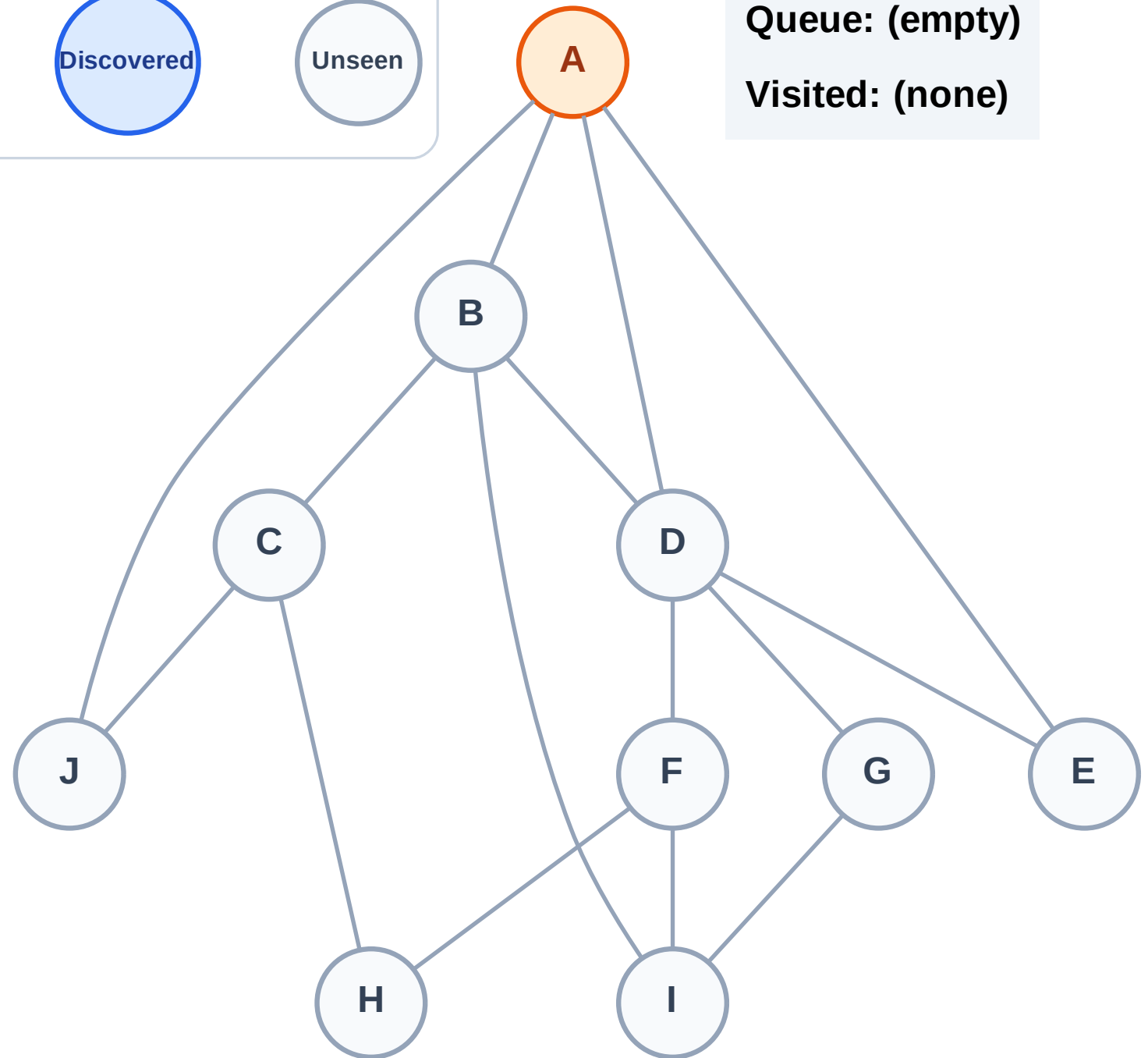
Processing

Discovered

Unseen

Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, D, E, J from A

Legend

Complete

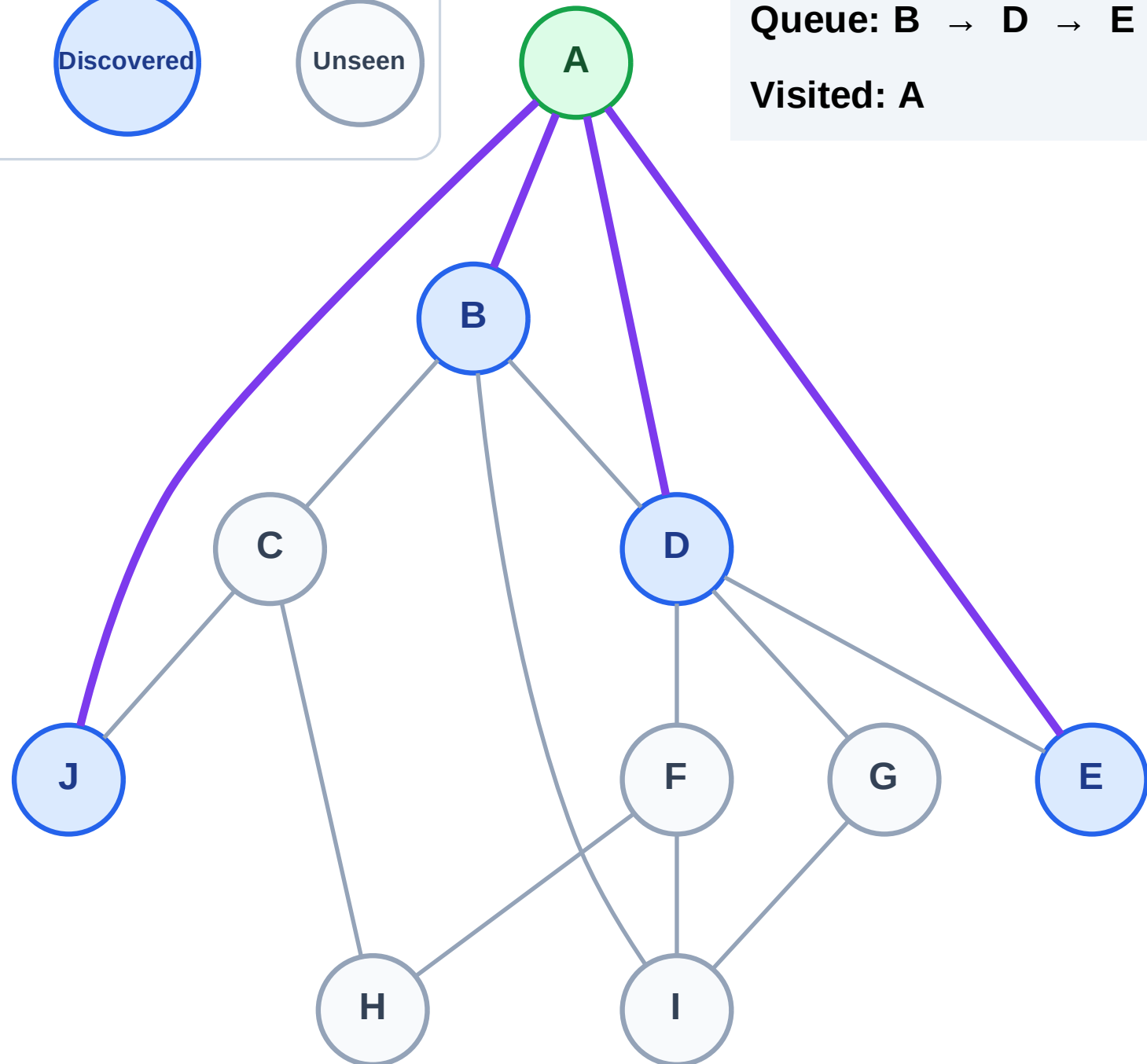
Processing

Discovered

Unseen

Queue: B → D → E → J

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

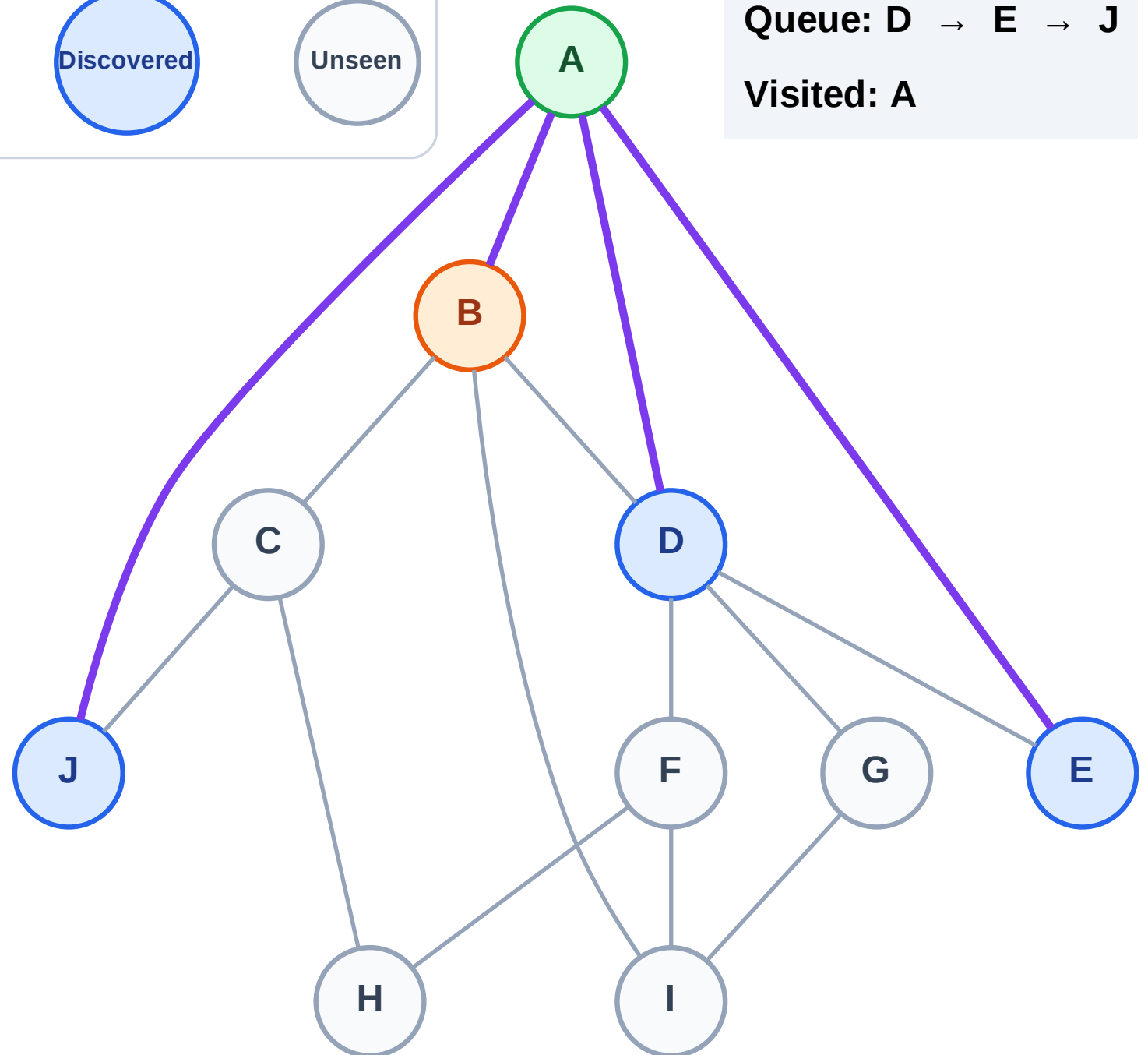
Processing

Discovered

Unseen

Queue: D → E → J

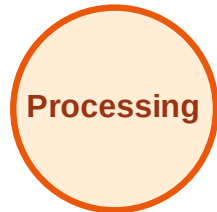
Visited: A



BFS Traversal

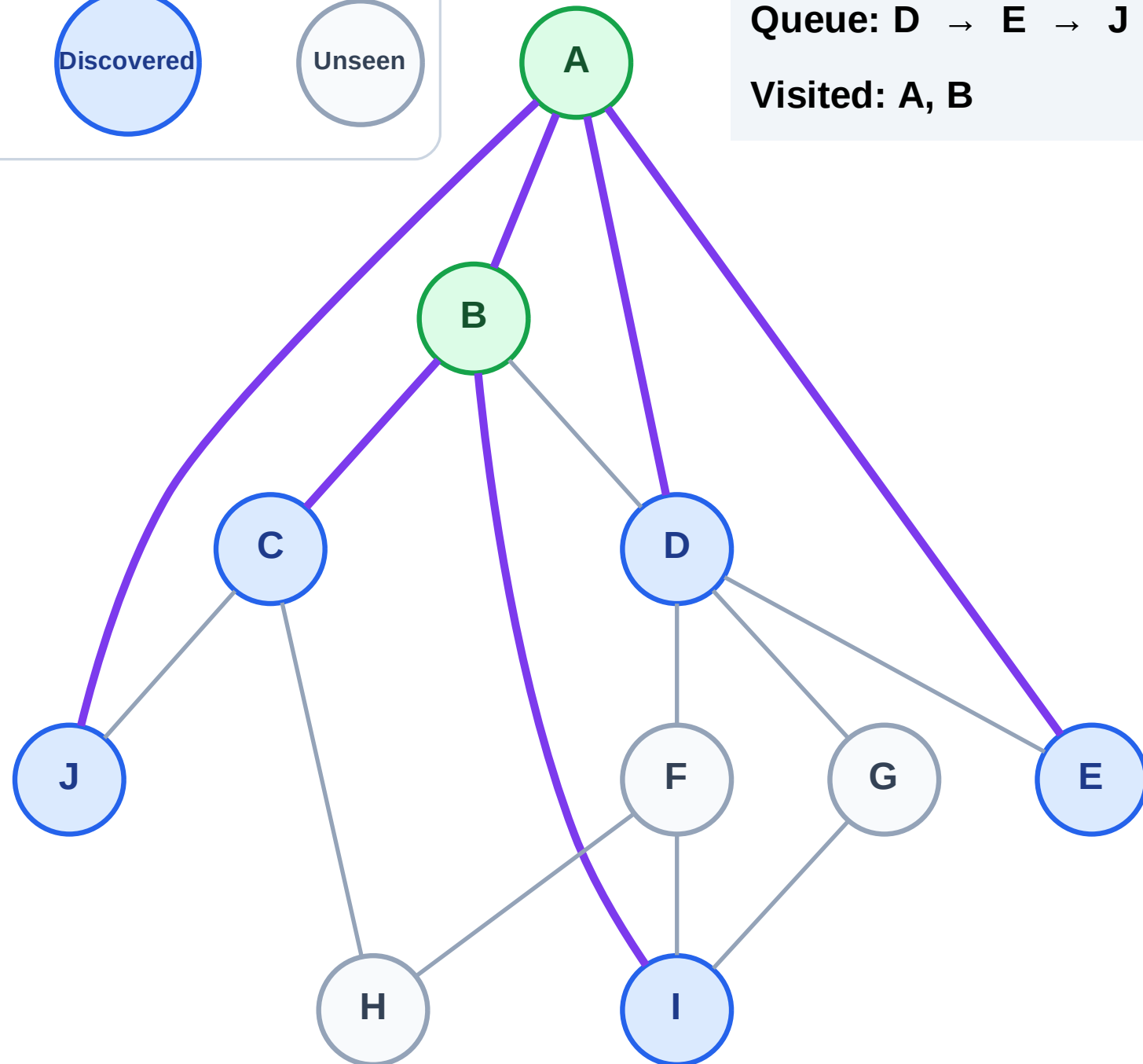
Step 5: Discover C, I from B

Legend



Queue: D → E → J → C → I

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

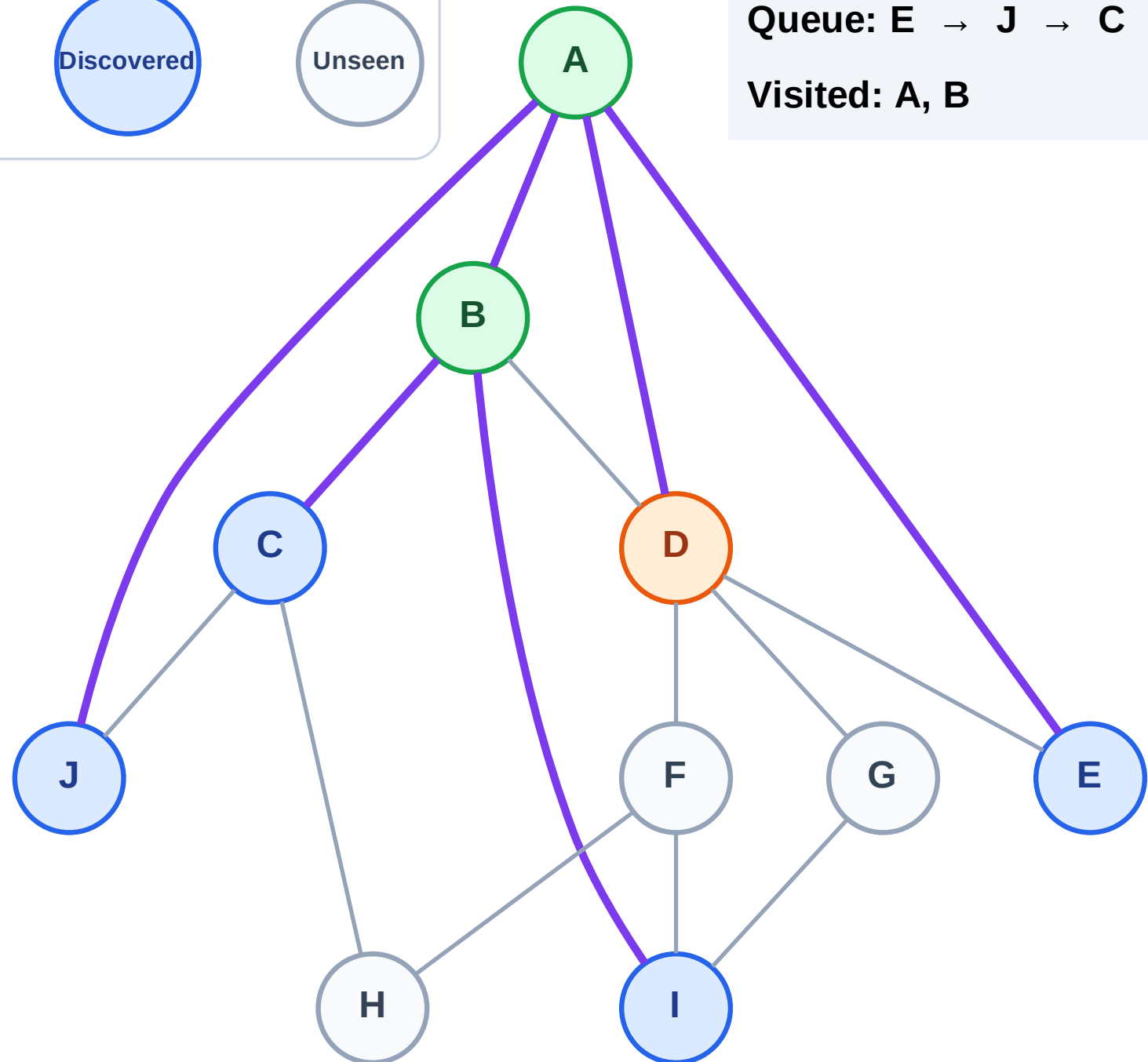
Processing

Discovered

Unseen

Queue: E → J → C → I

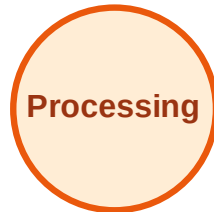
Visited: A, B



BFS Traversal

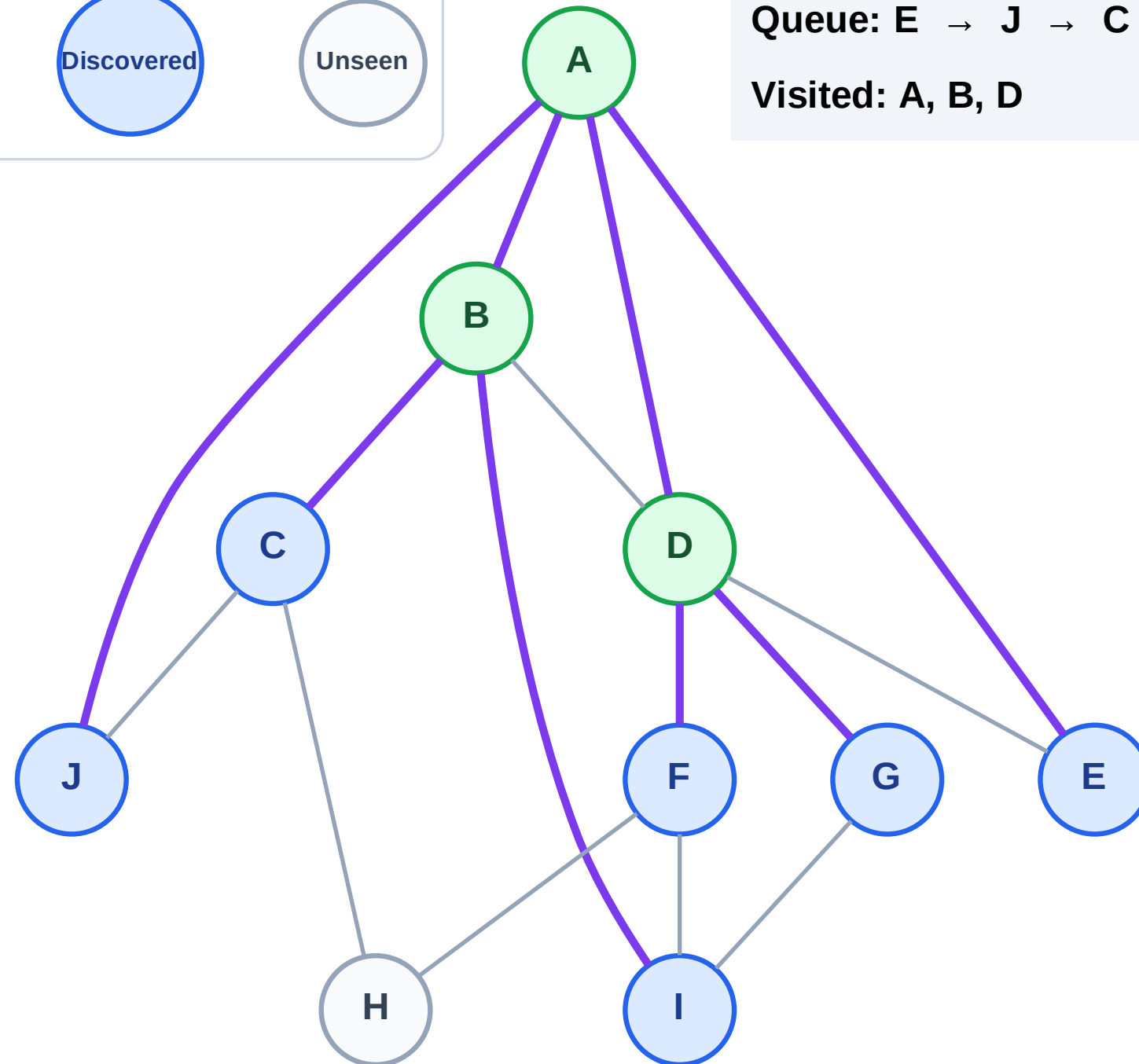
Step 7: Discover F, G from D

Legend



Queue: E → J → C → I → F → G

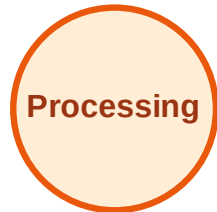
Visited: A, B, D



BFS Traversal

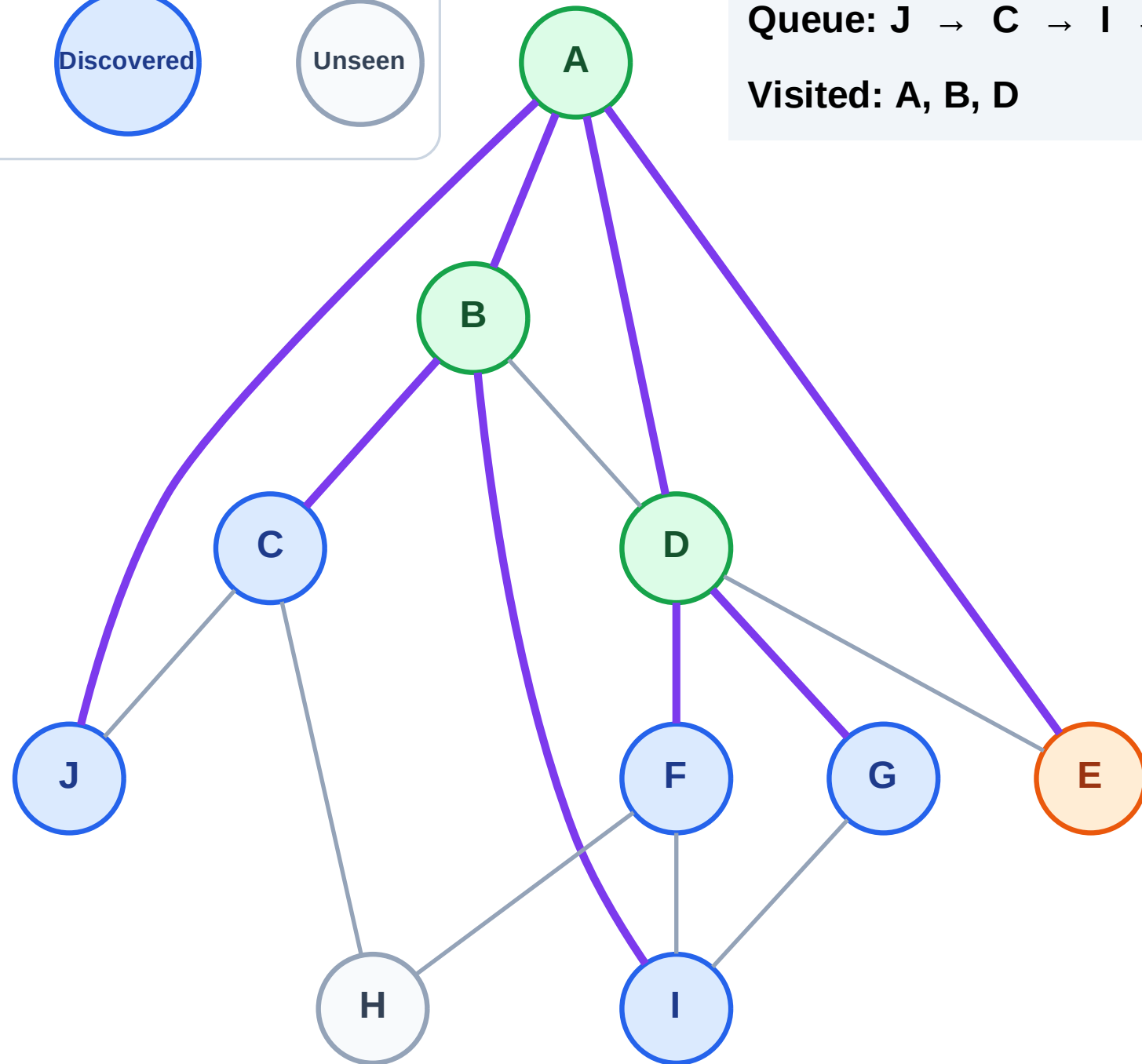
Step 8: Process node E

Legend



Queue: J → C → I → F → G

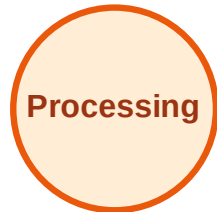
Visited: A, B, D



BFS Traversal

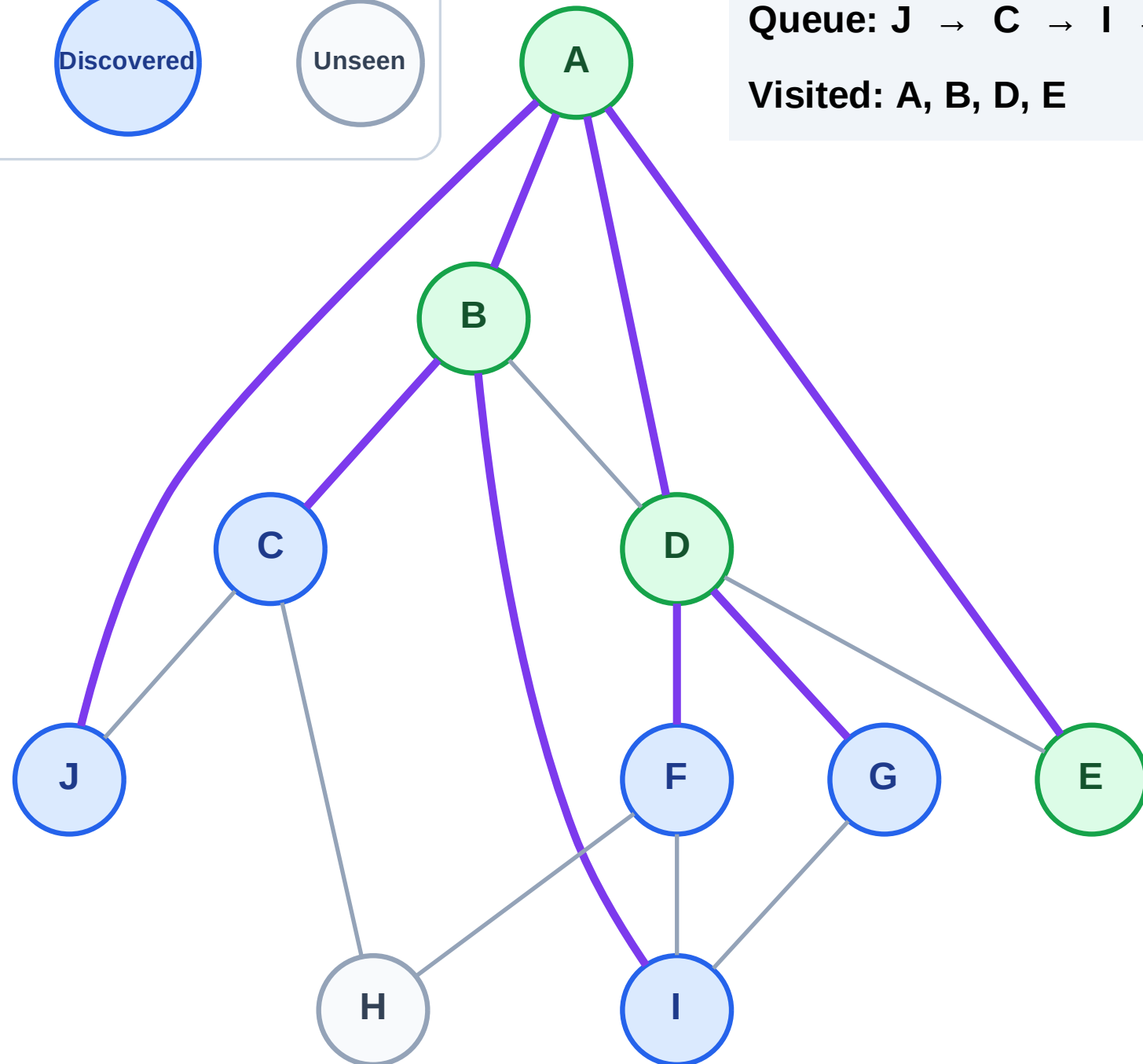
Step 9: No new neighbors from E

Legend



Queue: J → C → I → F → G

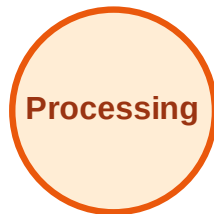
Visited: A, B, D, E



BFS Traversal

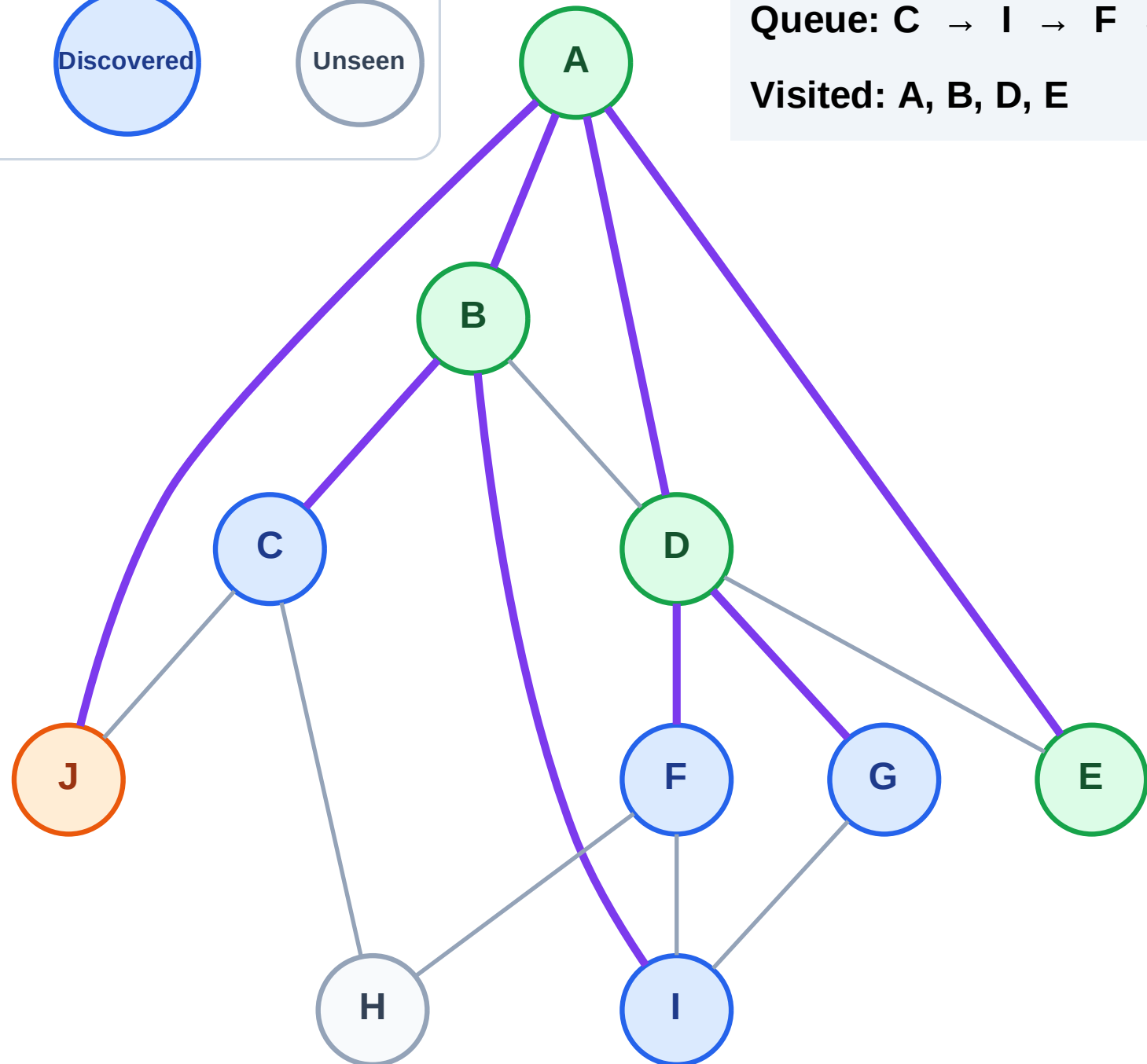
Step 10: Process node J

Legend



Queue: C → I → F → G

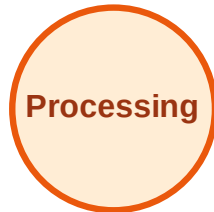
Visited: A, B, D, E



BFS Traversal

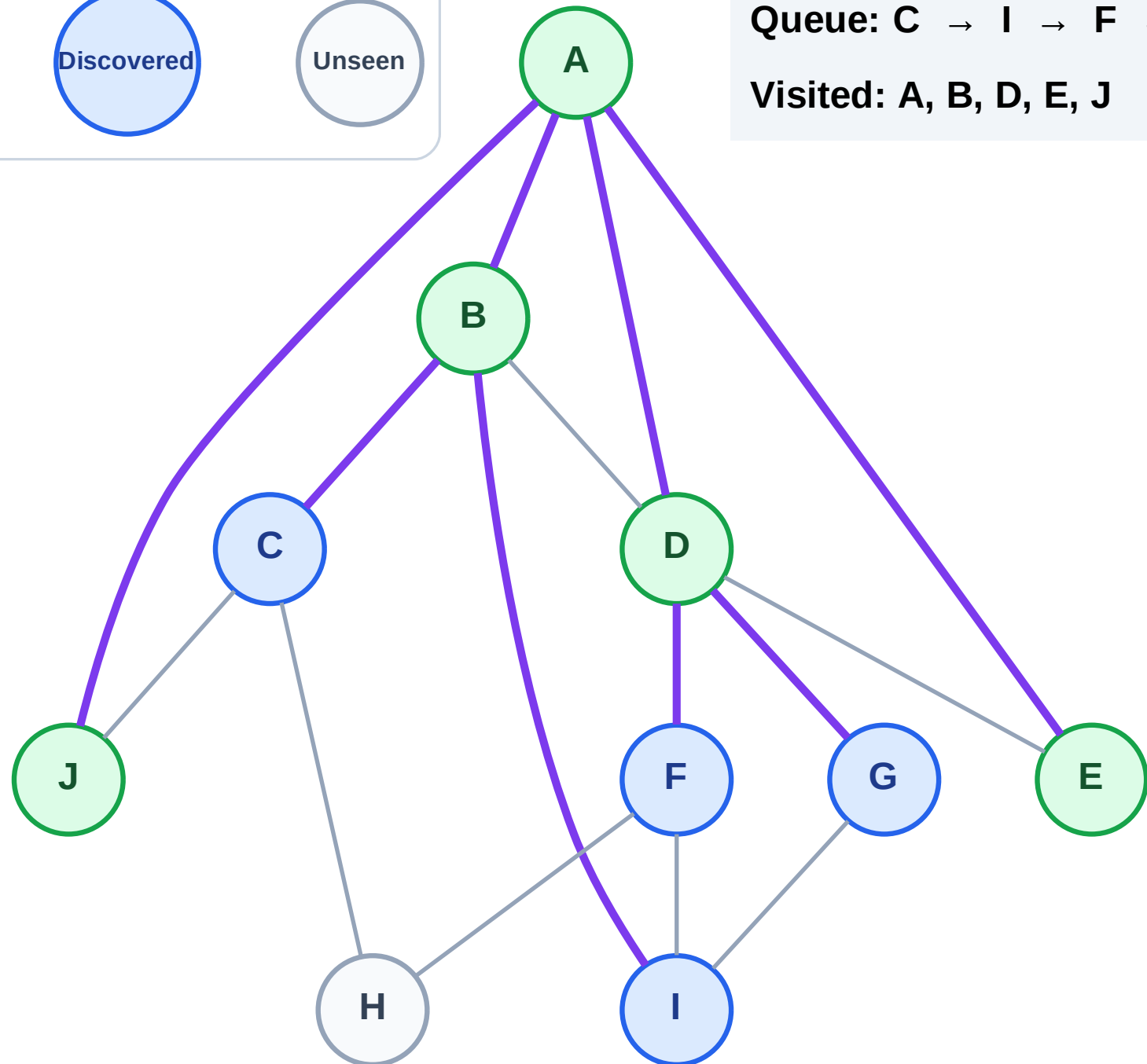
Step 11: No new neighbors from J

Legend



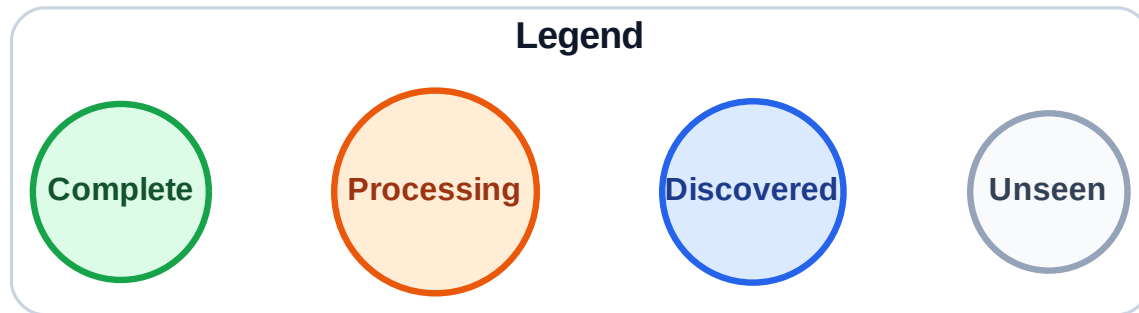
Queue: C → I → F → G

Visited: A, B, D, E, J

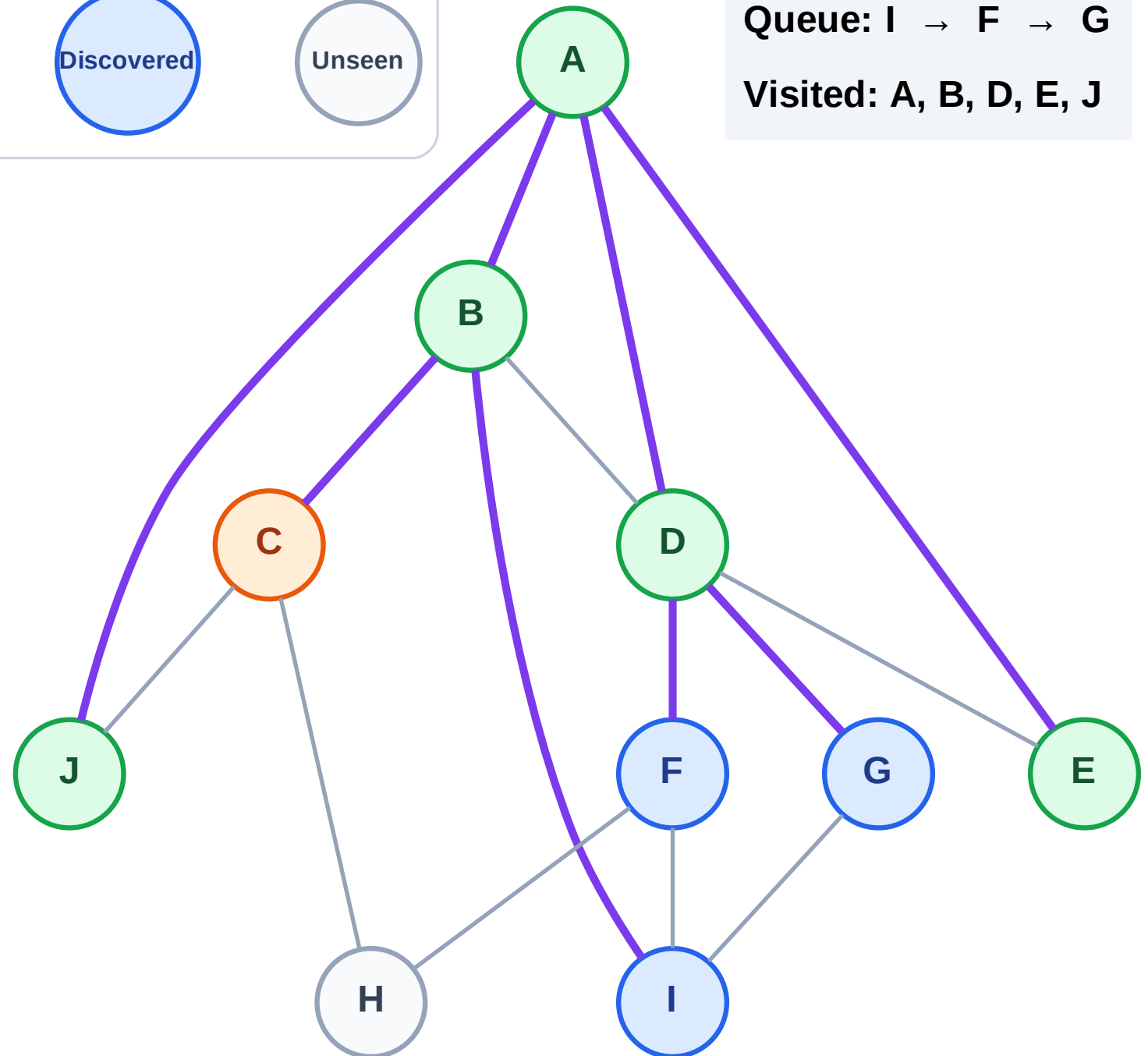


BFS Traversal

Step 12: Process node C



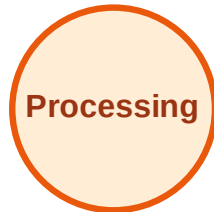
Queue: I → F → G
Visited: A, B, D, E, J



BFS Traversal

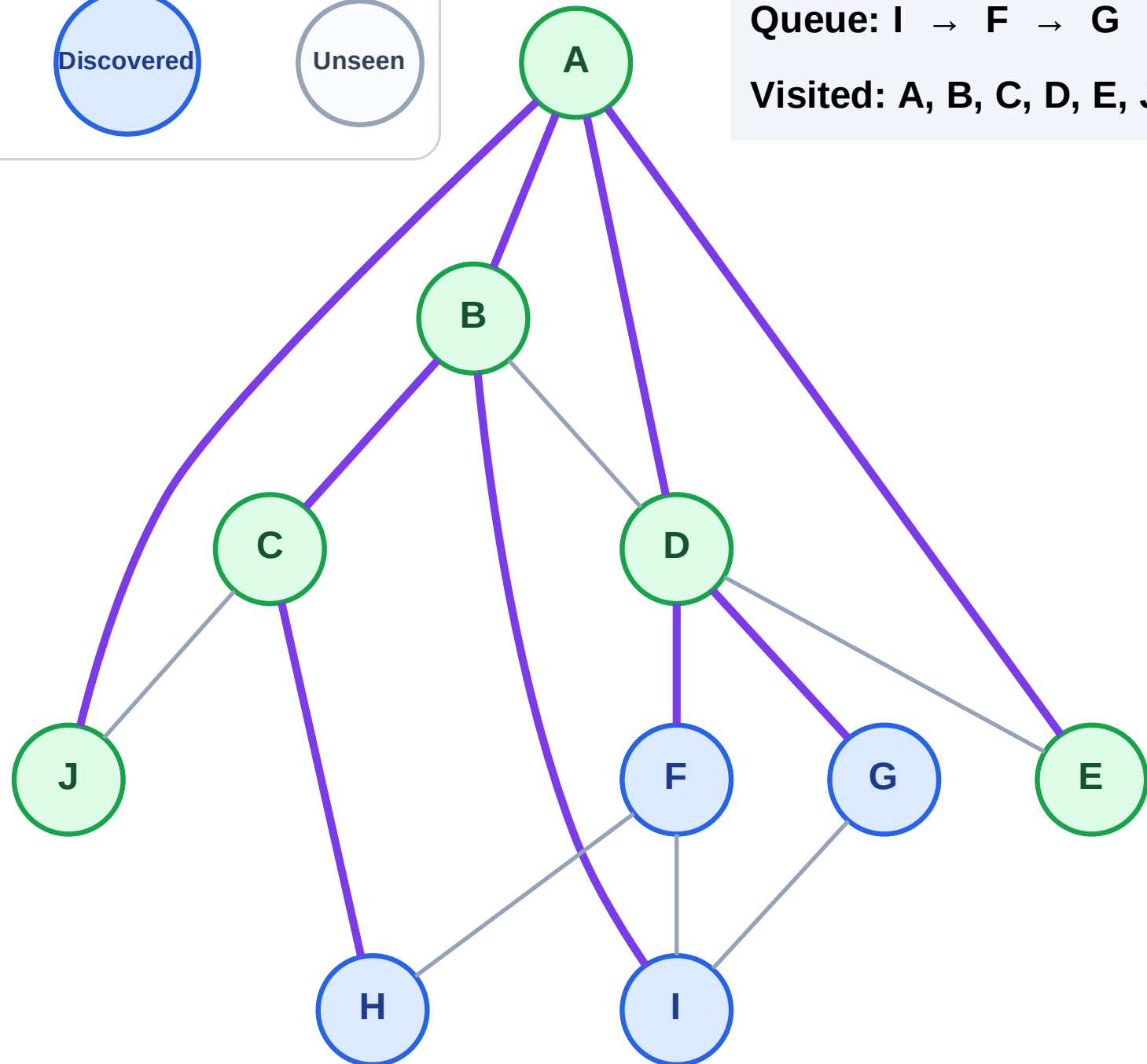
Step 13: Discover H from C

Legend



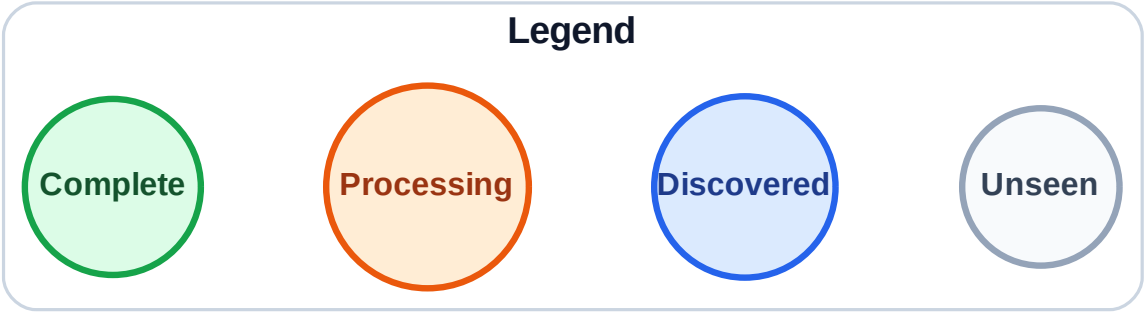
Queue: I → F → G → H

Visited: A, B, C, D, E, J

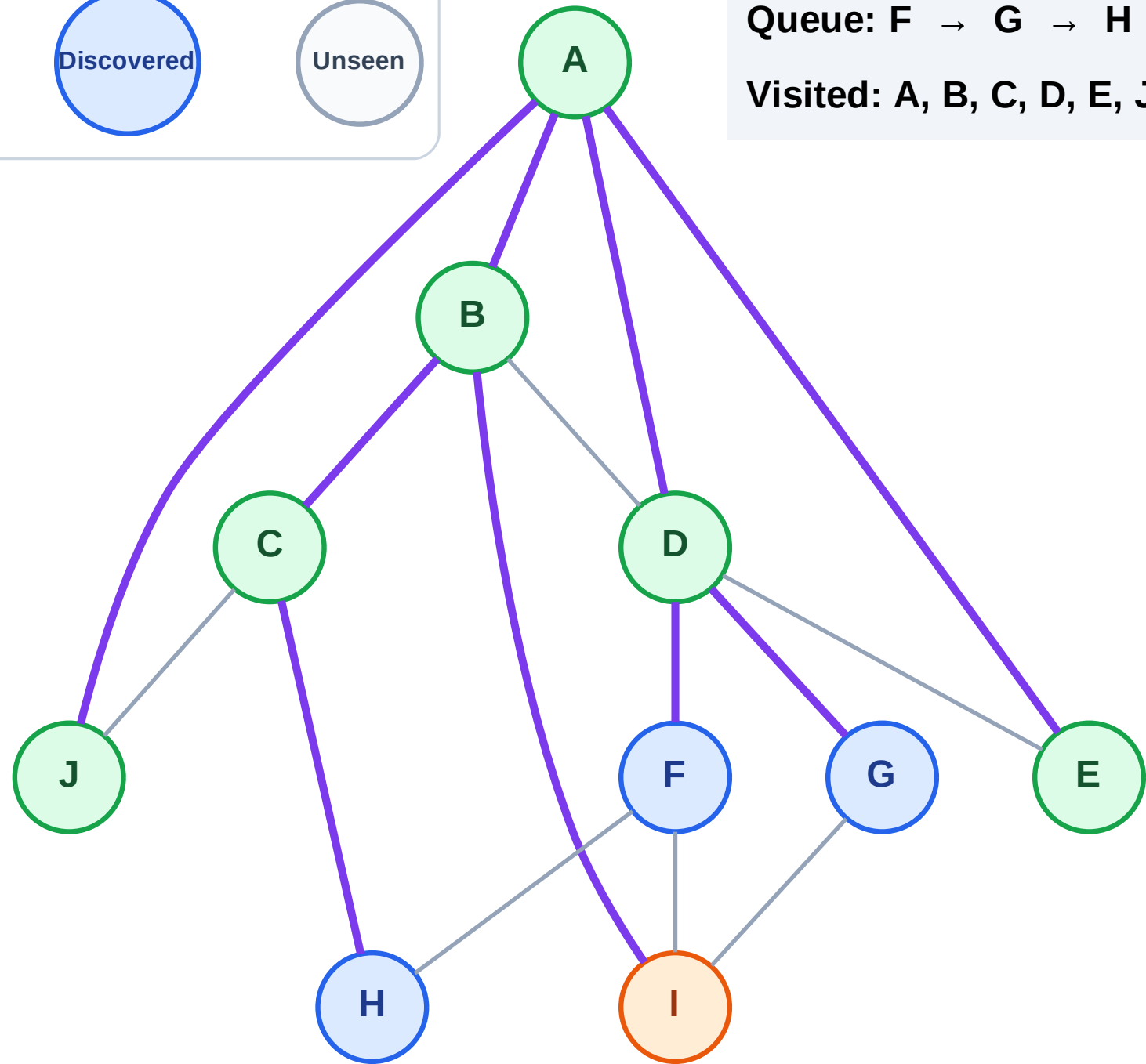


BFS Traversal

Step 14: Process node I



Queue: F → G → H
Visited: A, B, C, D, E, J



BFS Traversal

Step 15: No new neighbors from I

Legend

Complete

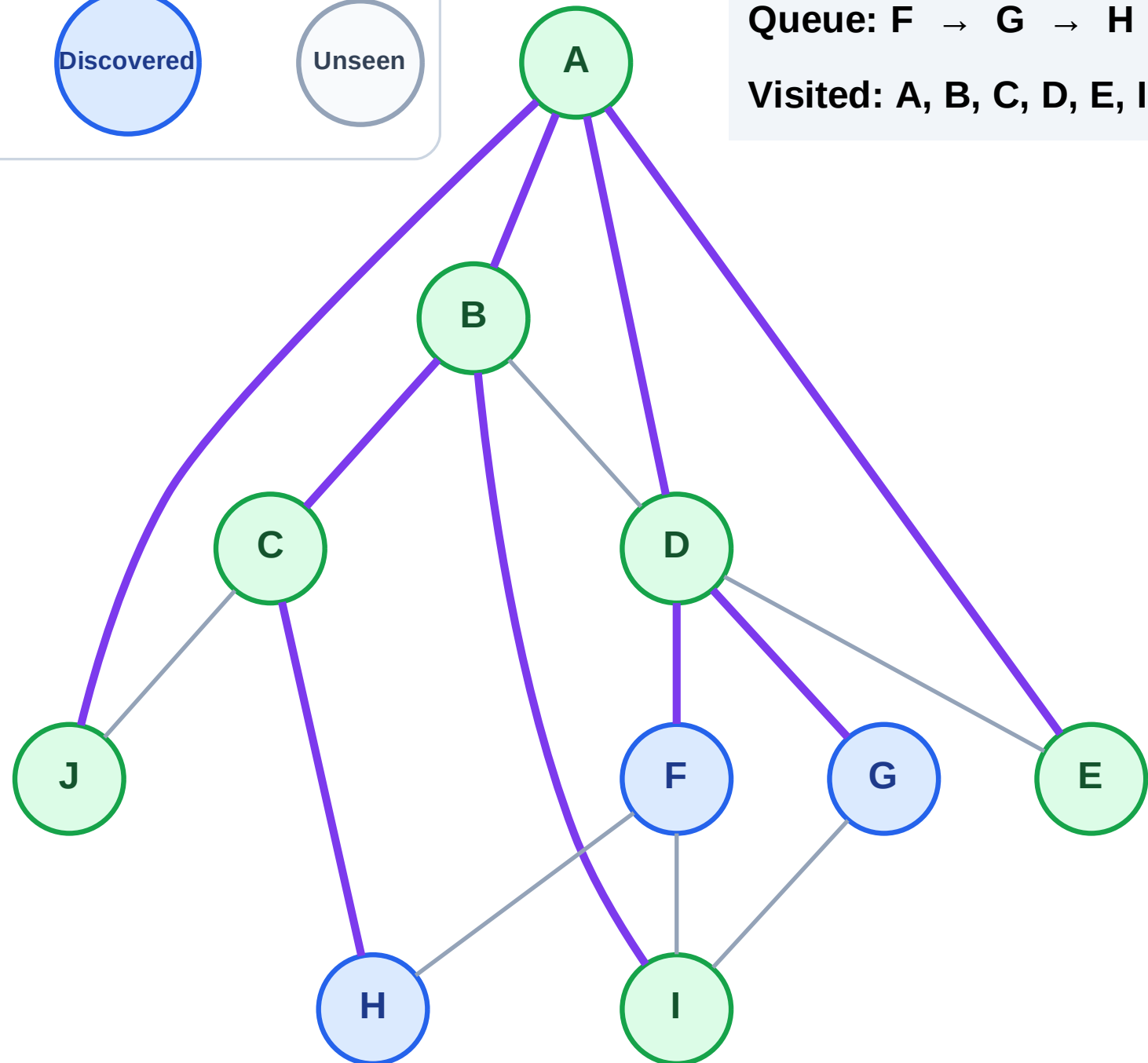
Processing

Discovered

Unseen

Queue: F → G → H

Visited: A, B, C, D, E, I, J



BFS Traversal

Step 16: Process node F

Legend

Complete

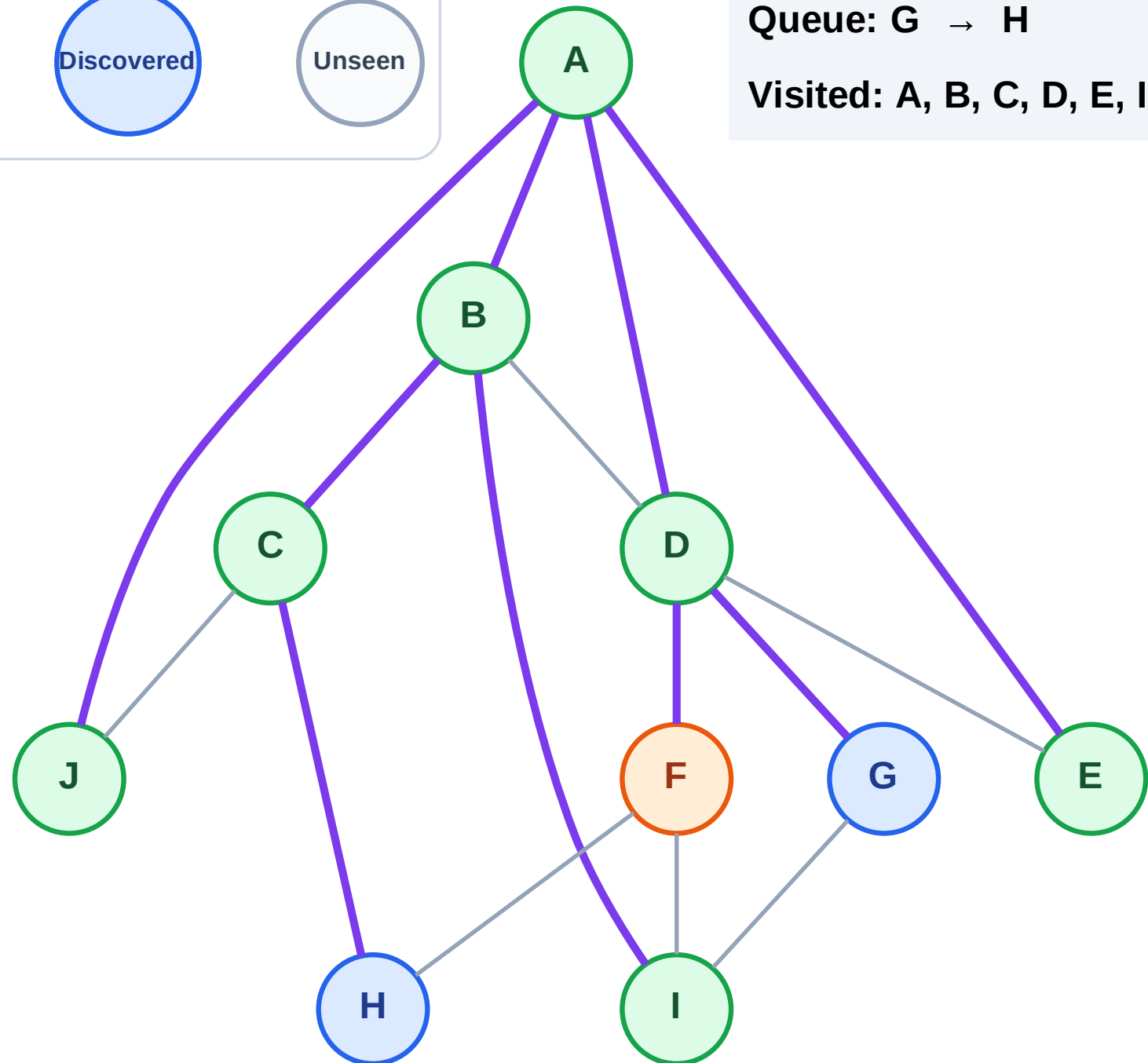
Processing

Discovered

Unseen

Queue: G → H

Visited: A, B, C, D, E, I, J



BFS Traversal

Step 17: No new neighbors from F

Legend

Complete

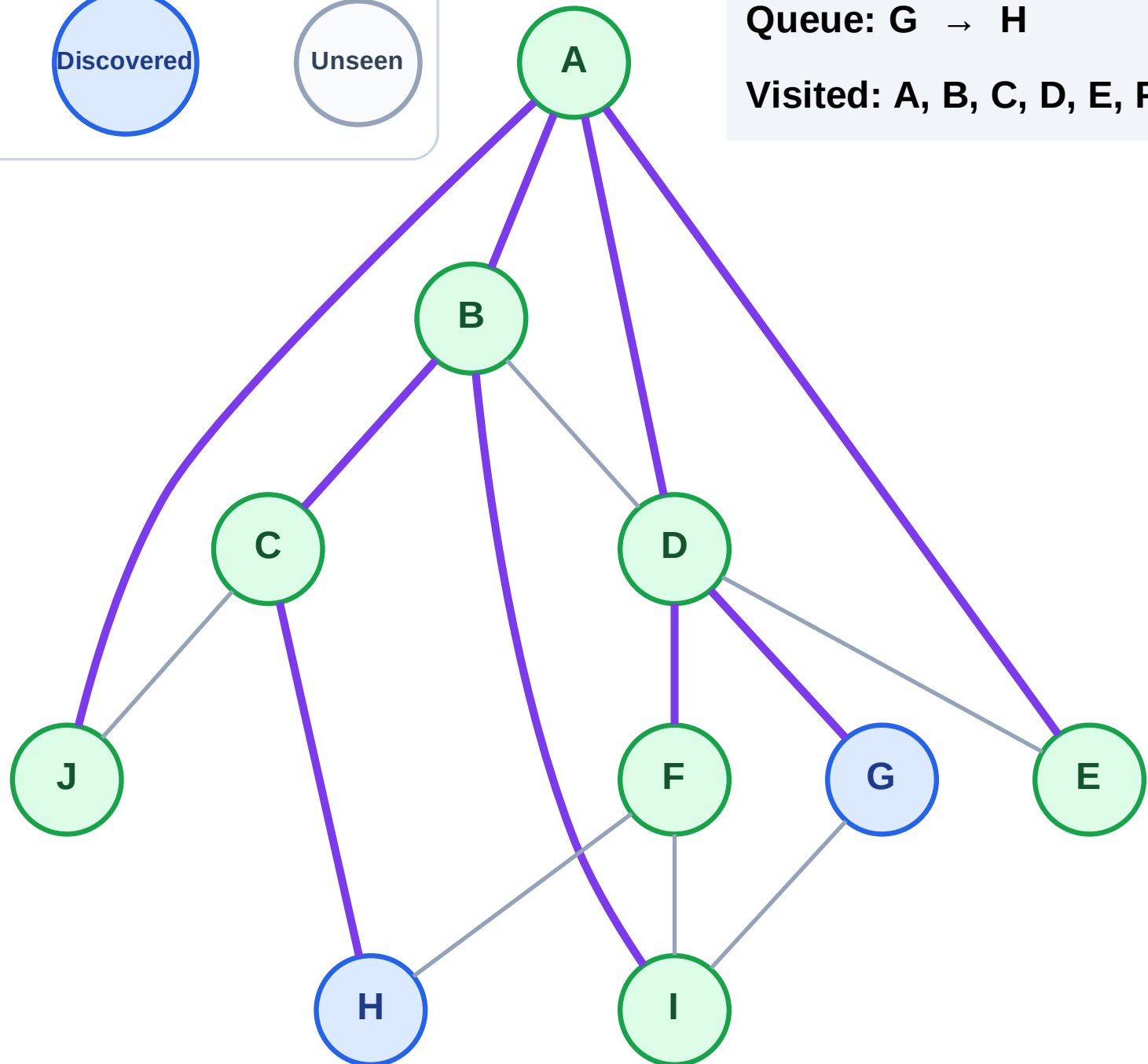
Processing

Discovered

Unseen

Queue: G → H

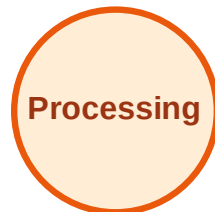
Visited: A, B, C, D, E, F, I, J



BFS Traversal

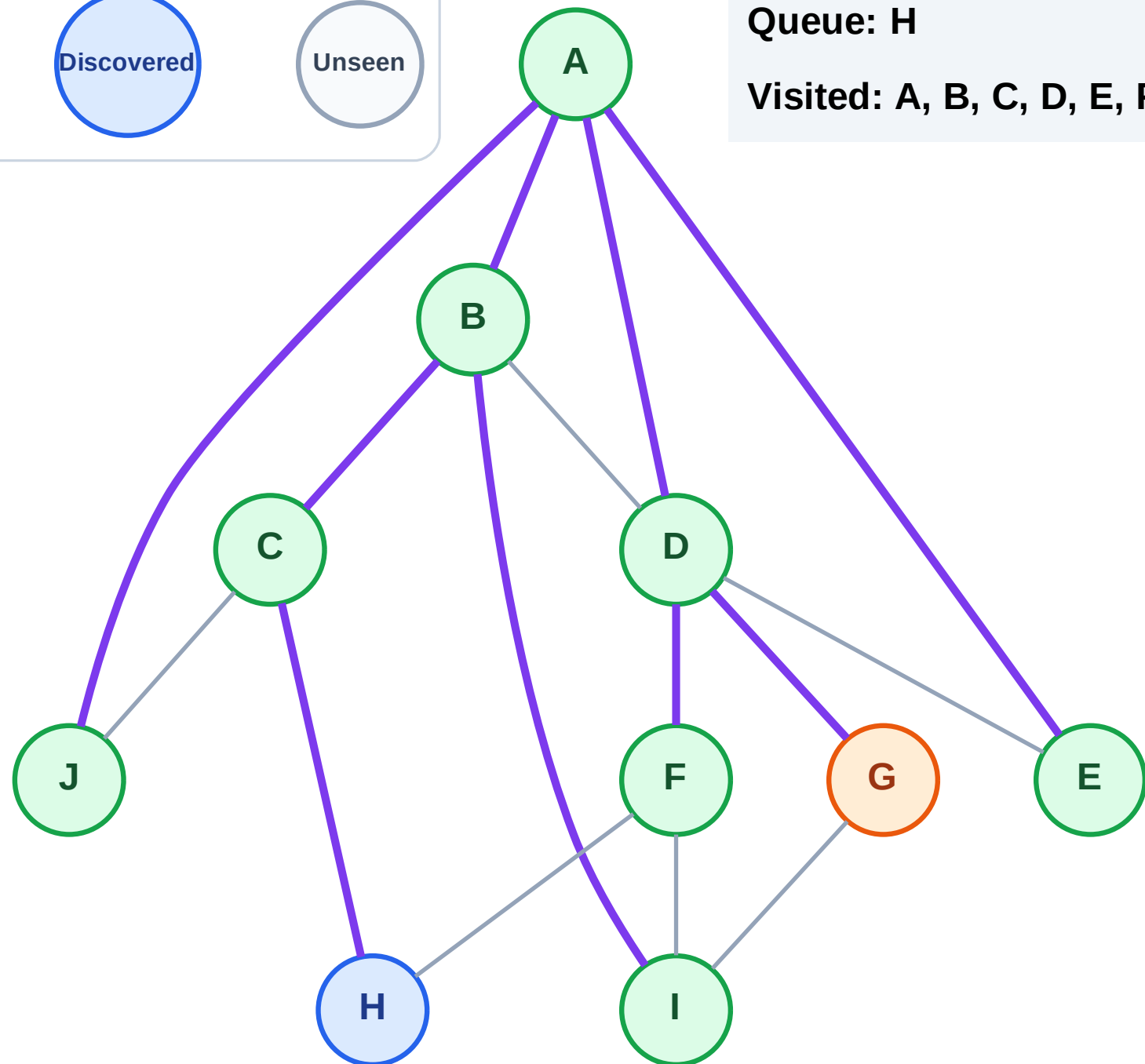
Step 18: Process node G

Legend



Queue: H

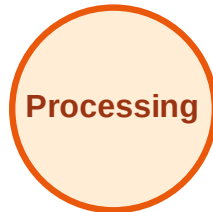
Visited: A, B, C, D, E, F, I, J



BFS Traversal

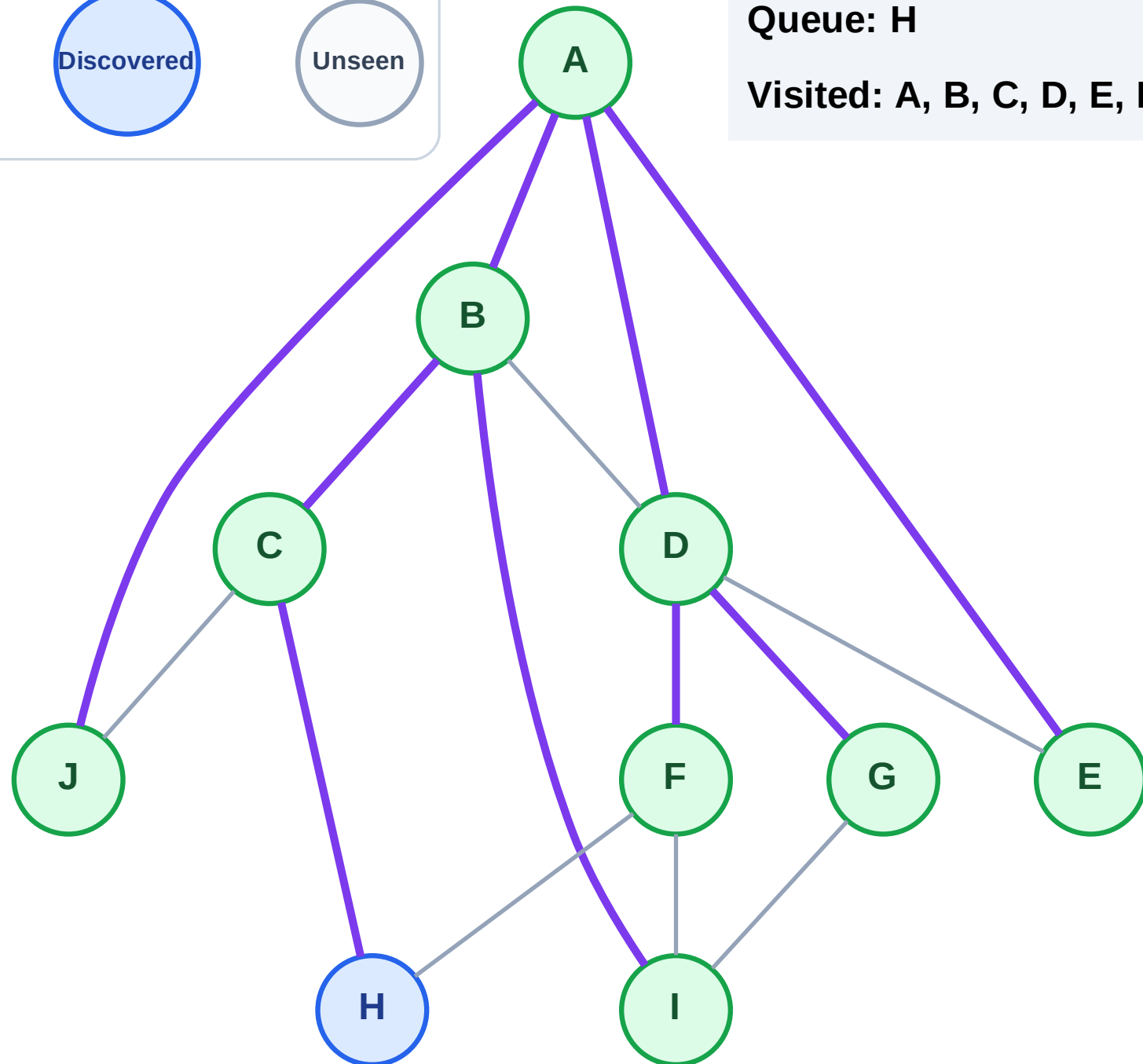
Step 19: No new neighbors from G

Legend



Queue: H

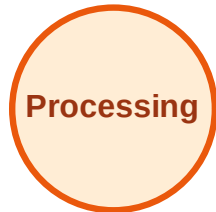
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

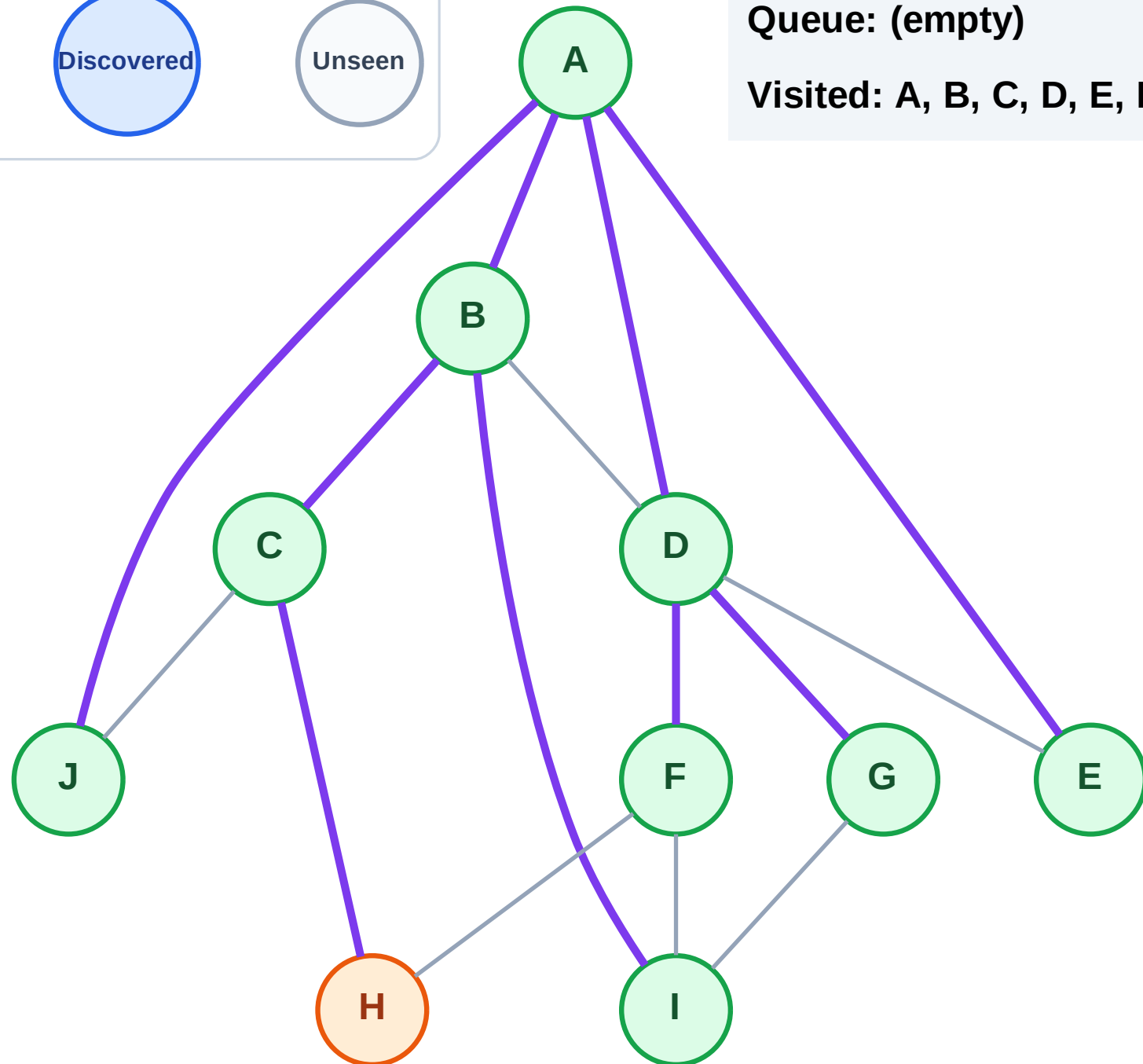
Step 20: Process node H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

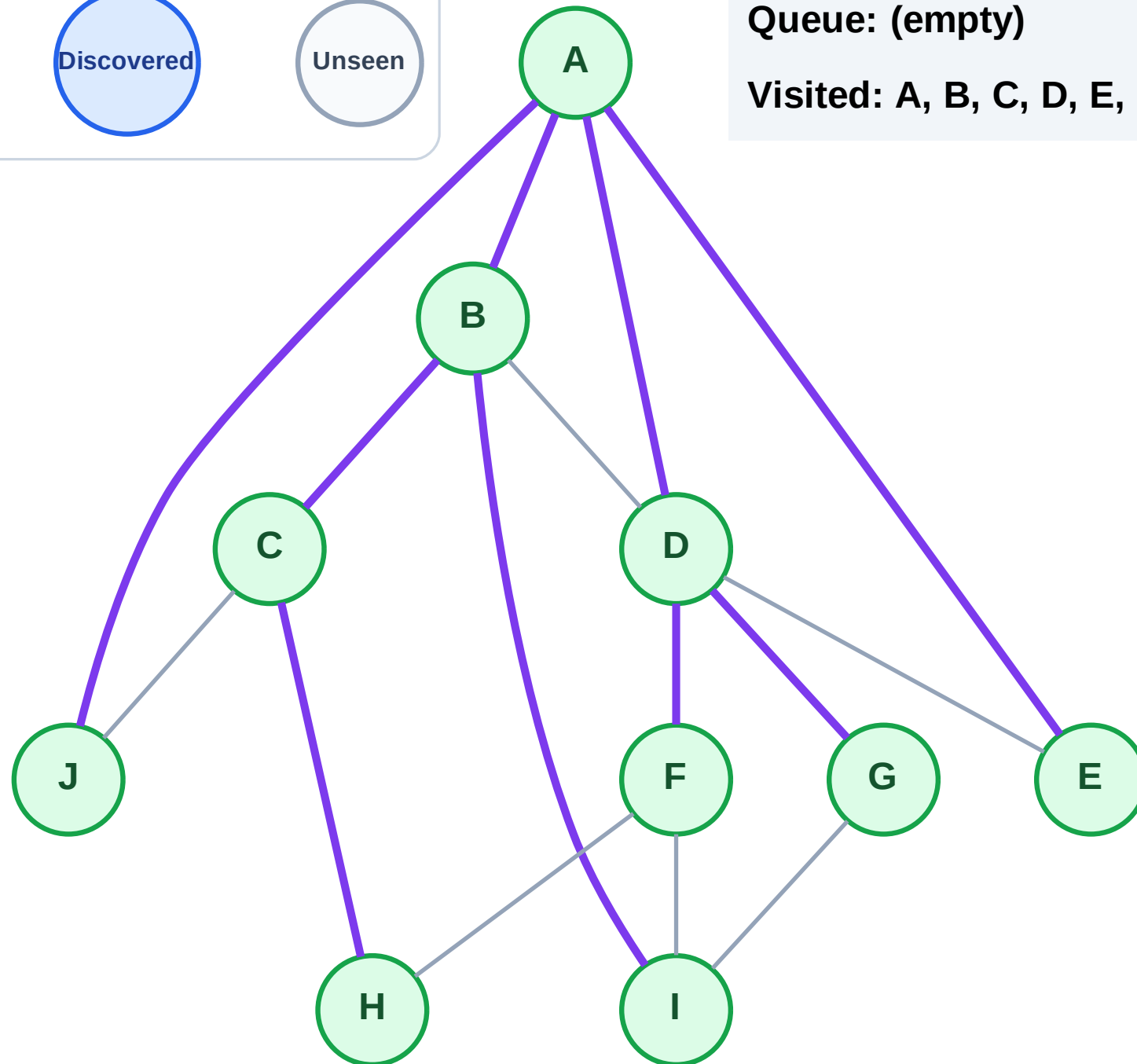
Step 21: No new neighbors from H

Legend



Queue: (empty)

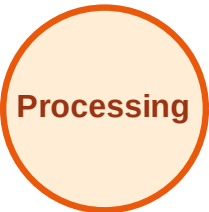
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

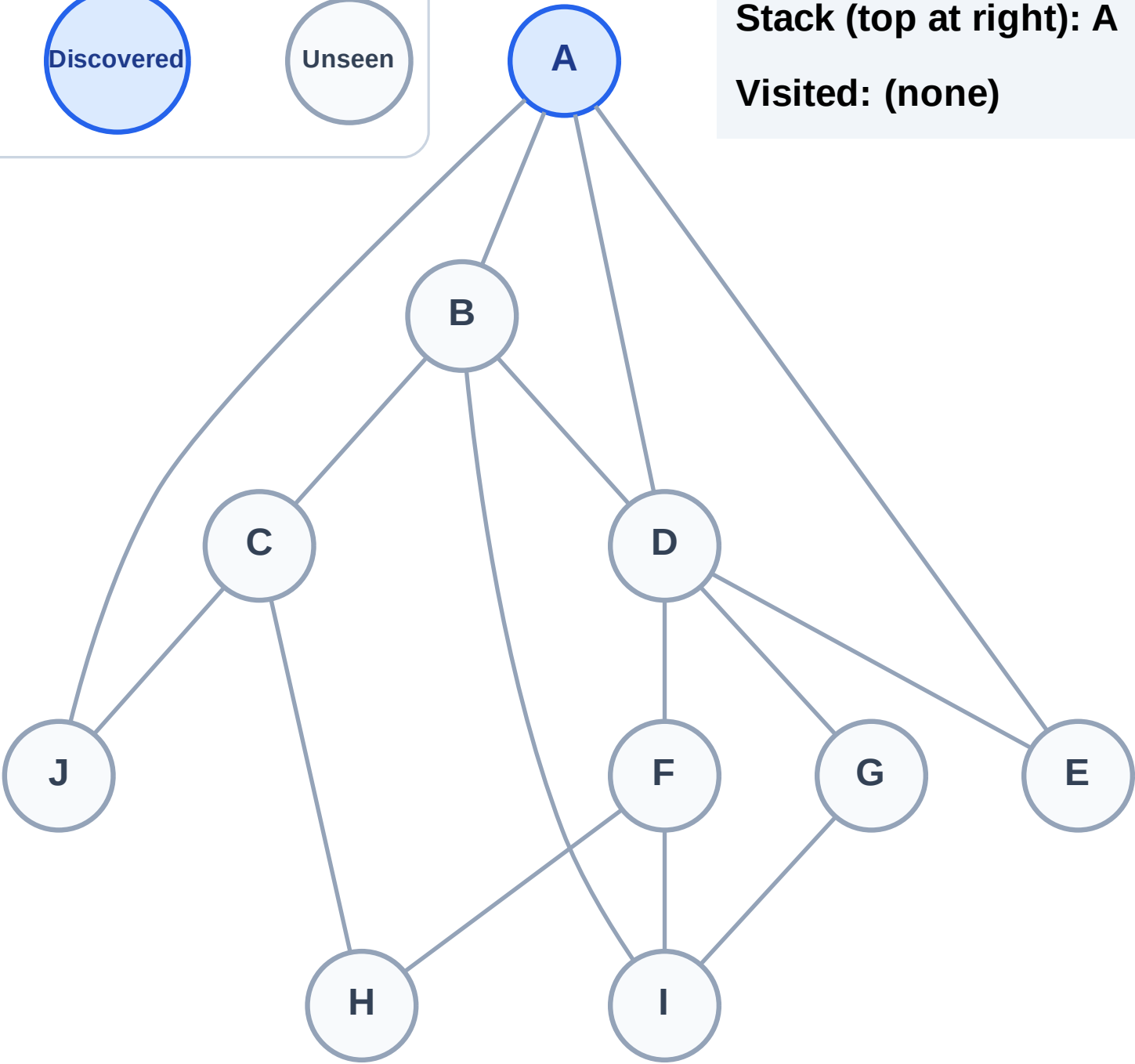
Step 1: Start at A

Legend



Stack (top at right): A

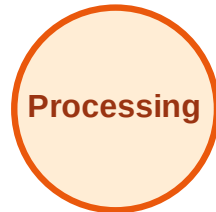
Visited: (none)



DFS Traversal

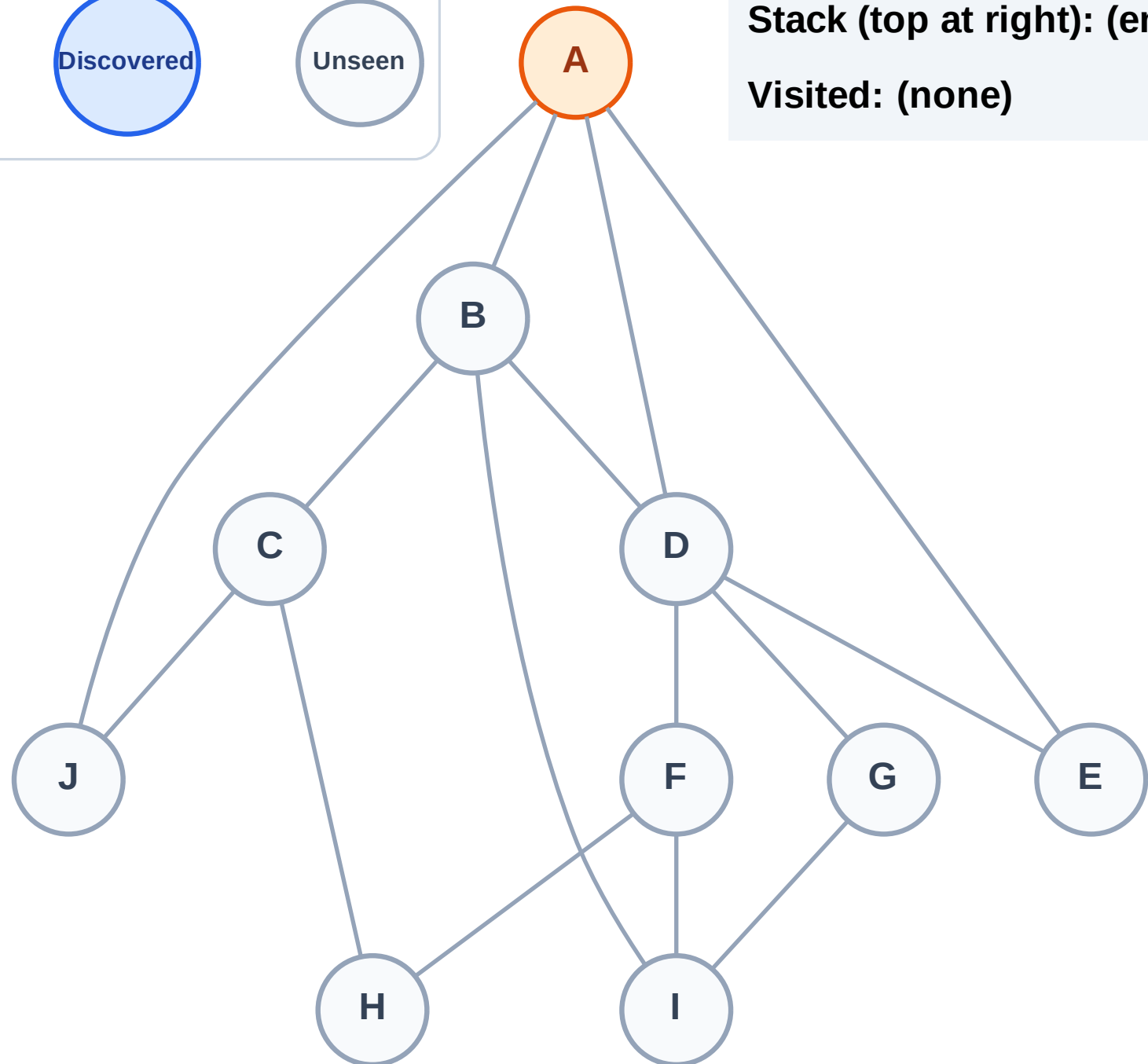
Step 2: Process node A

Legend



Stack (top at right): (empty)

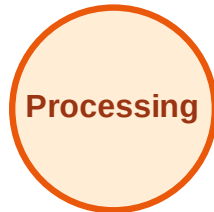
Visited: (none)



DFS Traversal

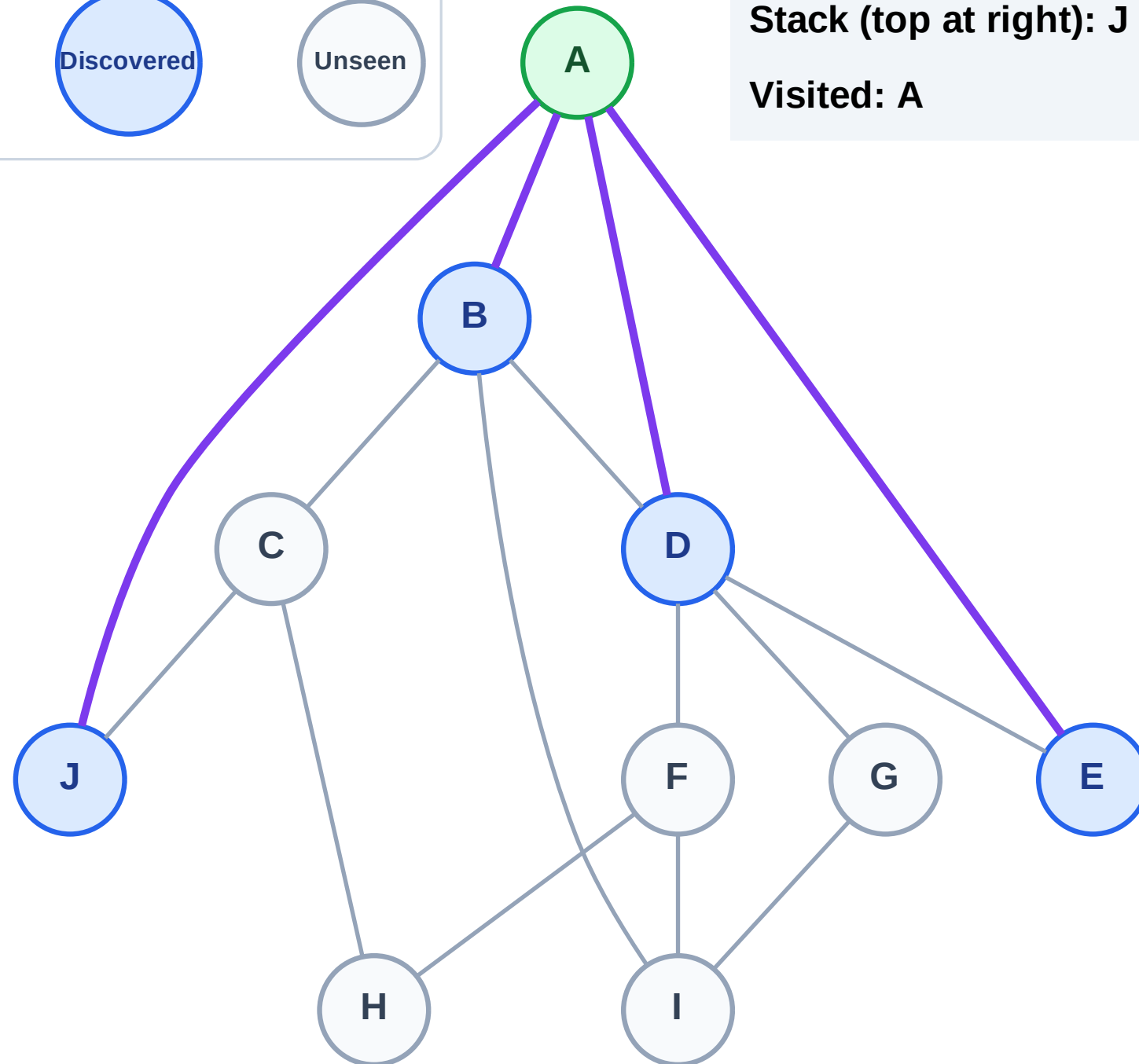
Step 3: Discover J, E, D, B from A

Legend



Stack (top at right): J | E | D | B

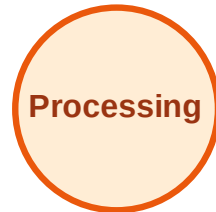
Visited: A



DFS Traversal

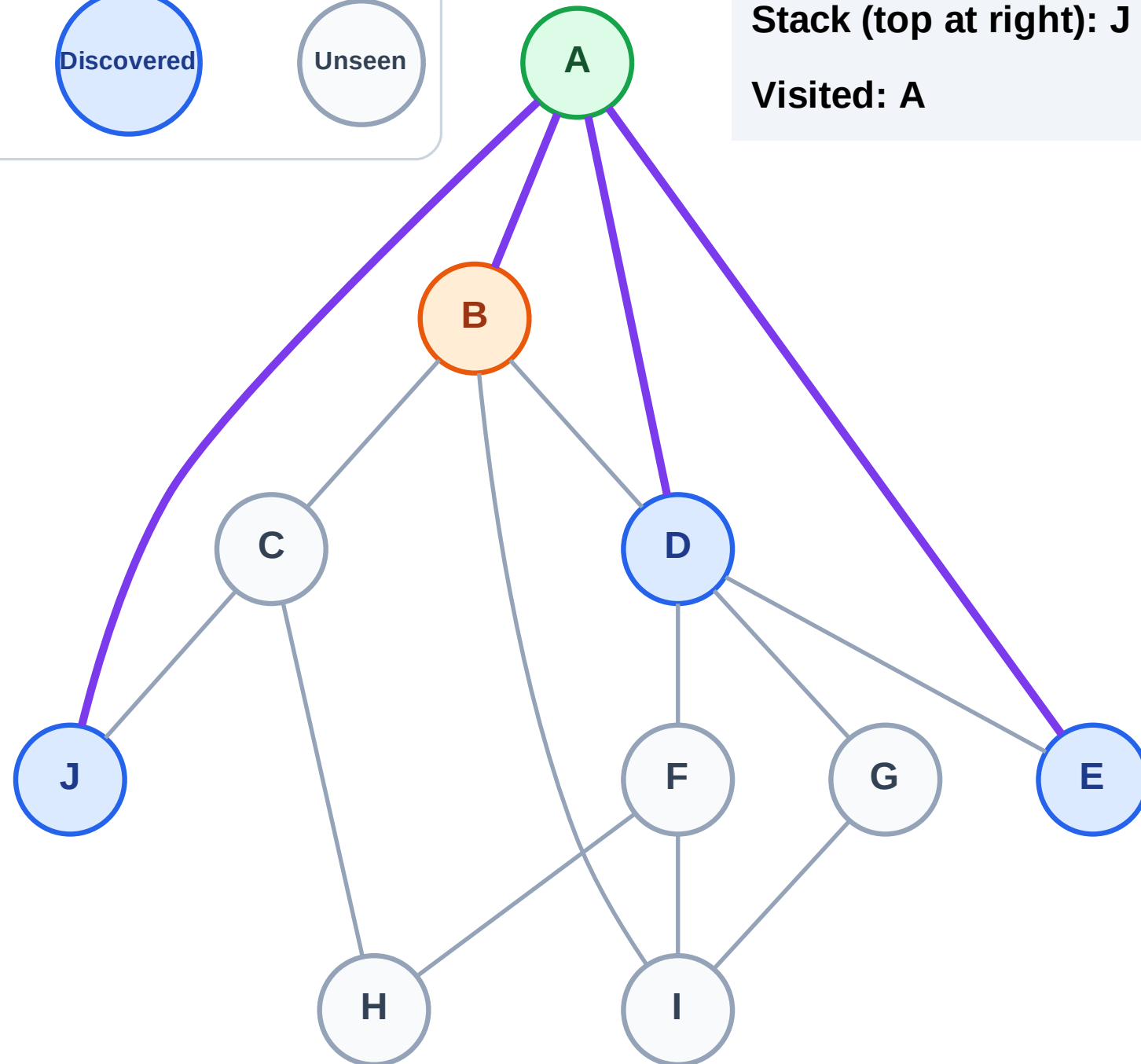
Step 4: Process node B

Legend



Stack (top at right): J | E | D

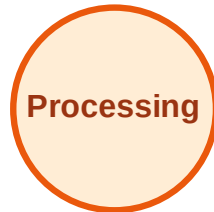
Visited: A



DFS Traversal

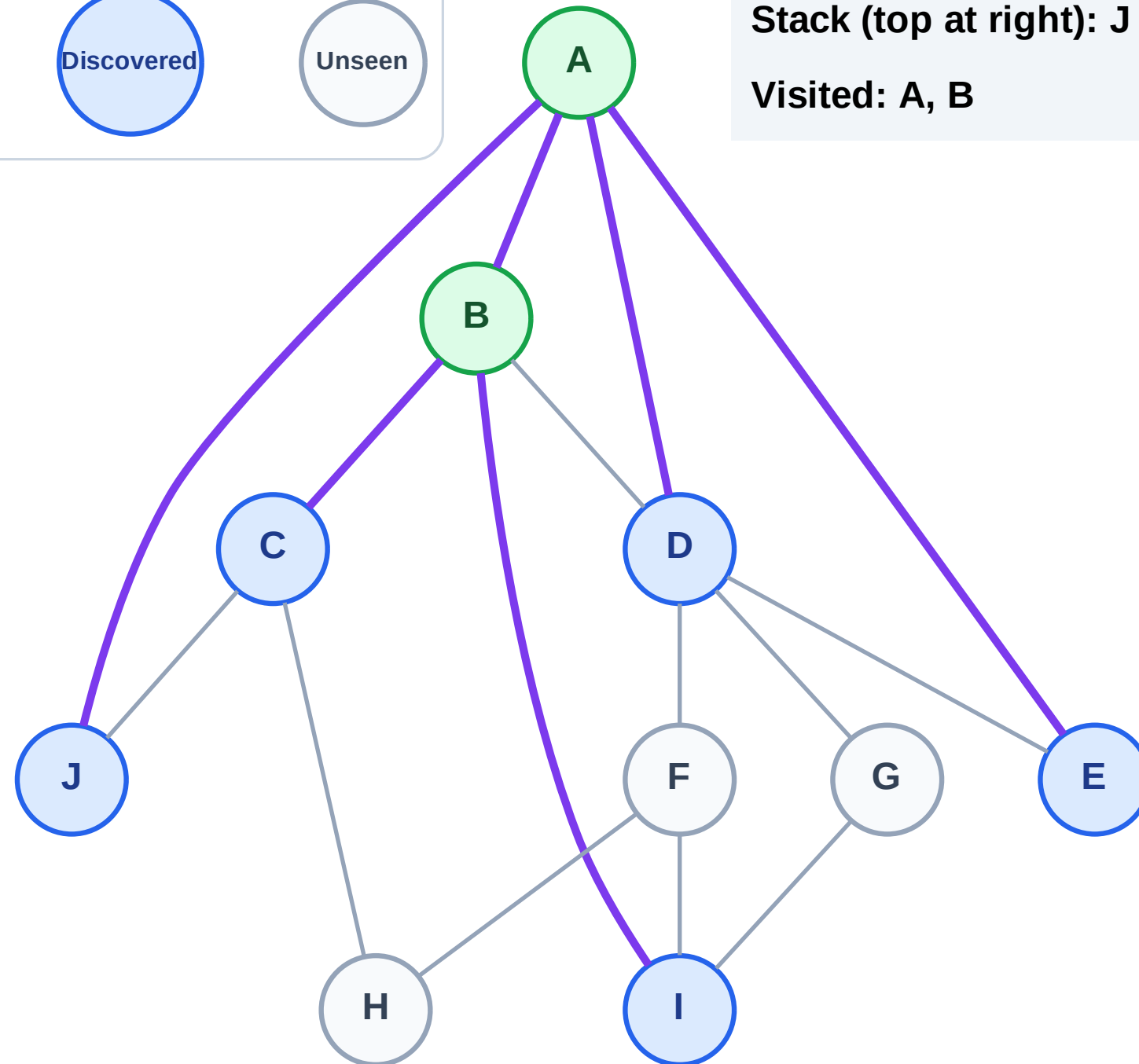
Step 5: Discover I, C from B

Legend



Stack (top at right): J | E | D | I | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

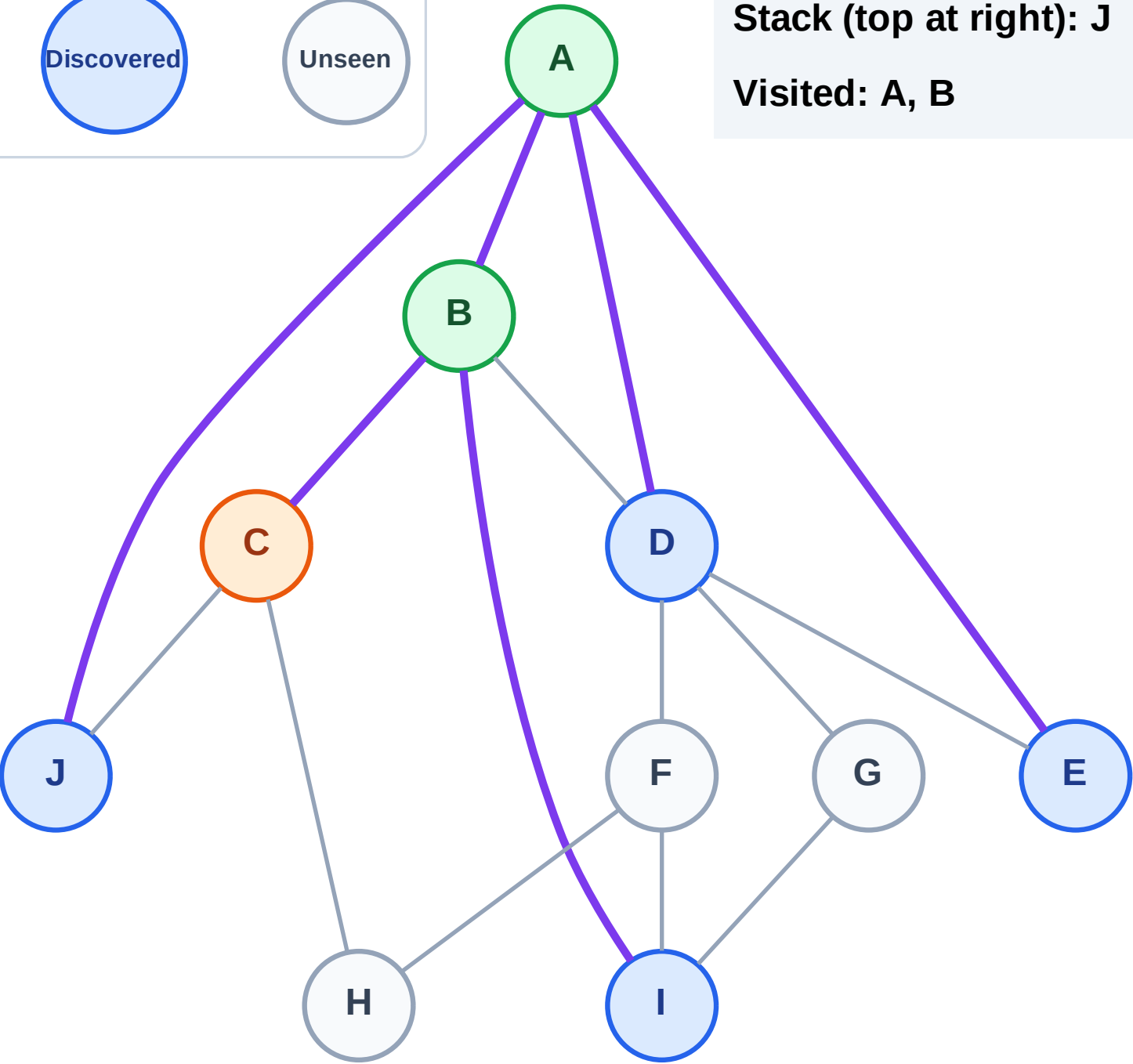
Processing

Discovered

Unseen

Stack (top at right): J | E | D | I

Visited: A, B



DFS Traversal

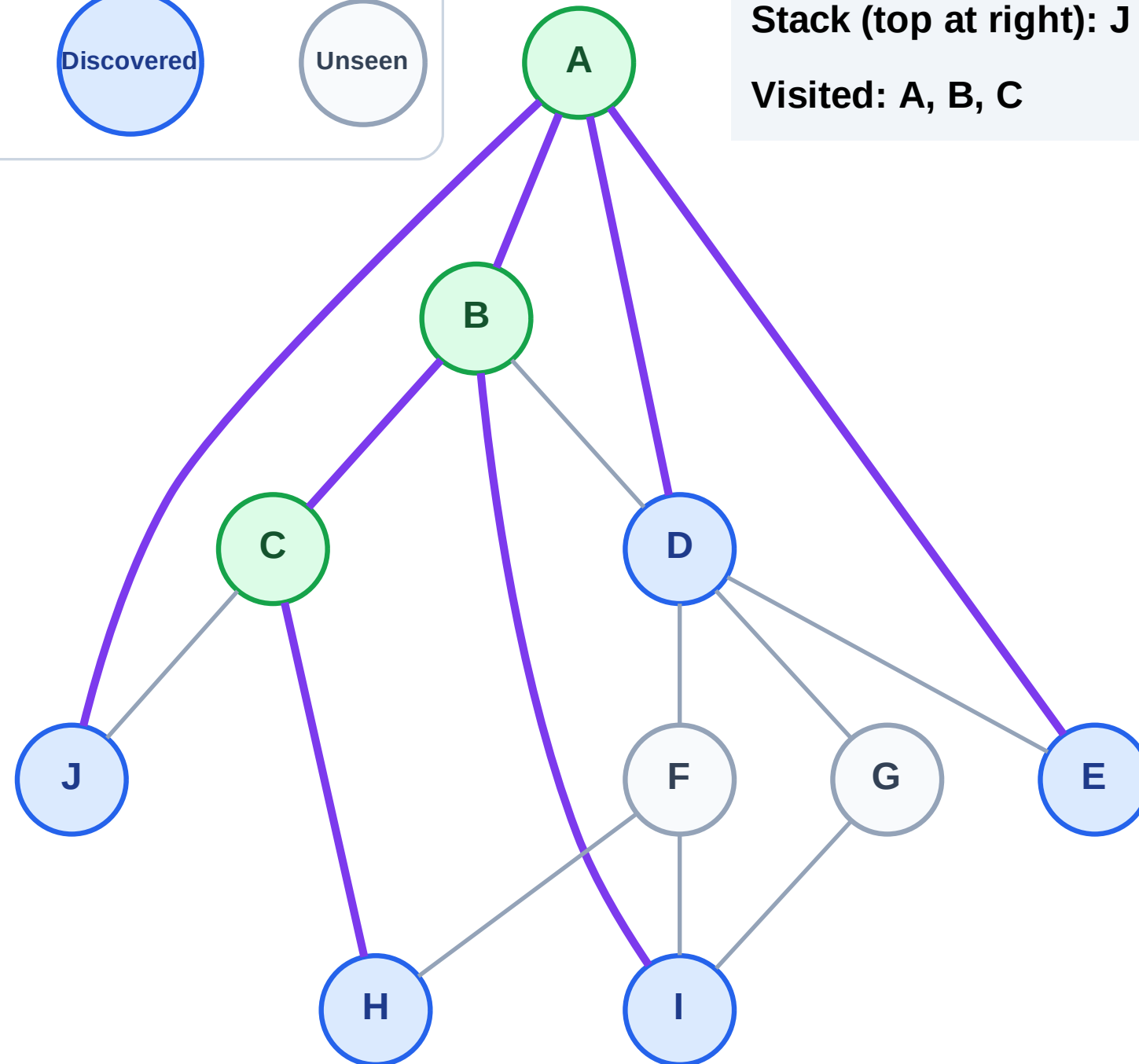
Step 7: Discover H from C

Legend



Stack (top at right): J | E | D | I | H

Visited: A, B, C



DFS Traversal

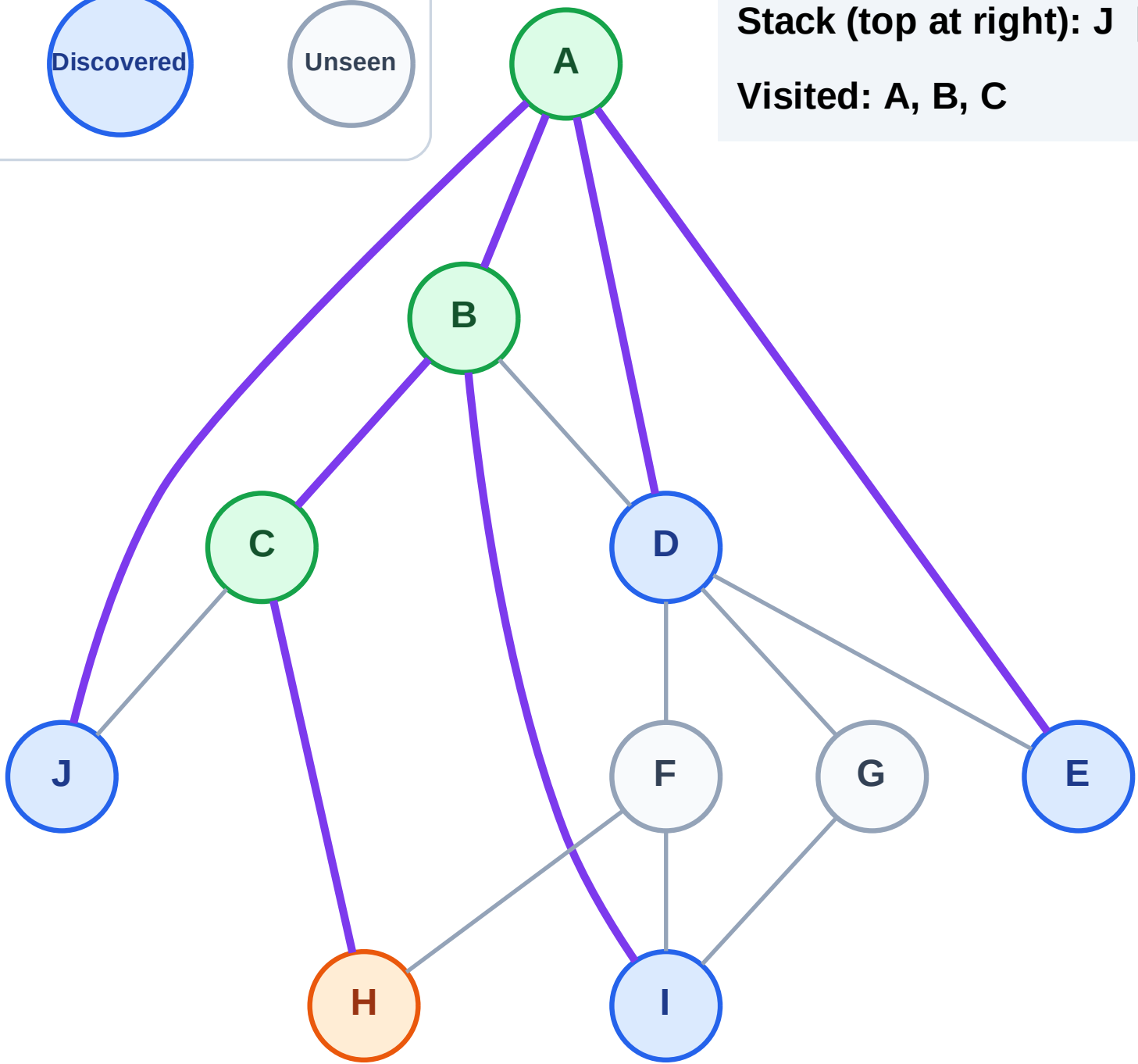
Step 8: Process node H

Legend



Stack (top at right): J | E | D | I

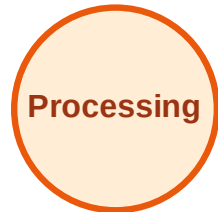
Visited: A, B, C



DFS Traversal

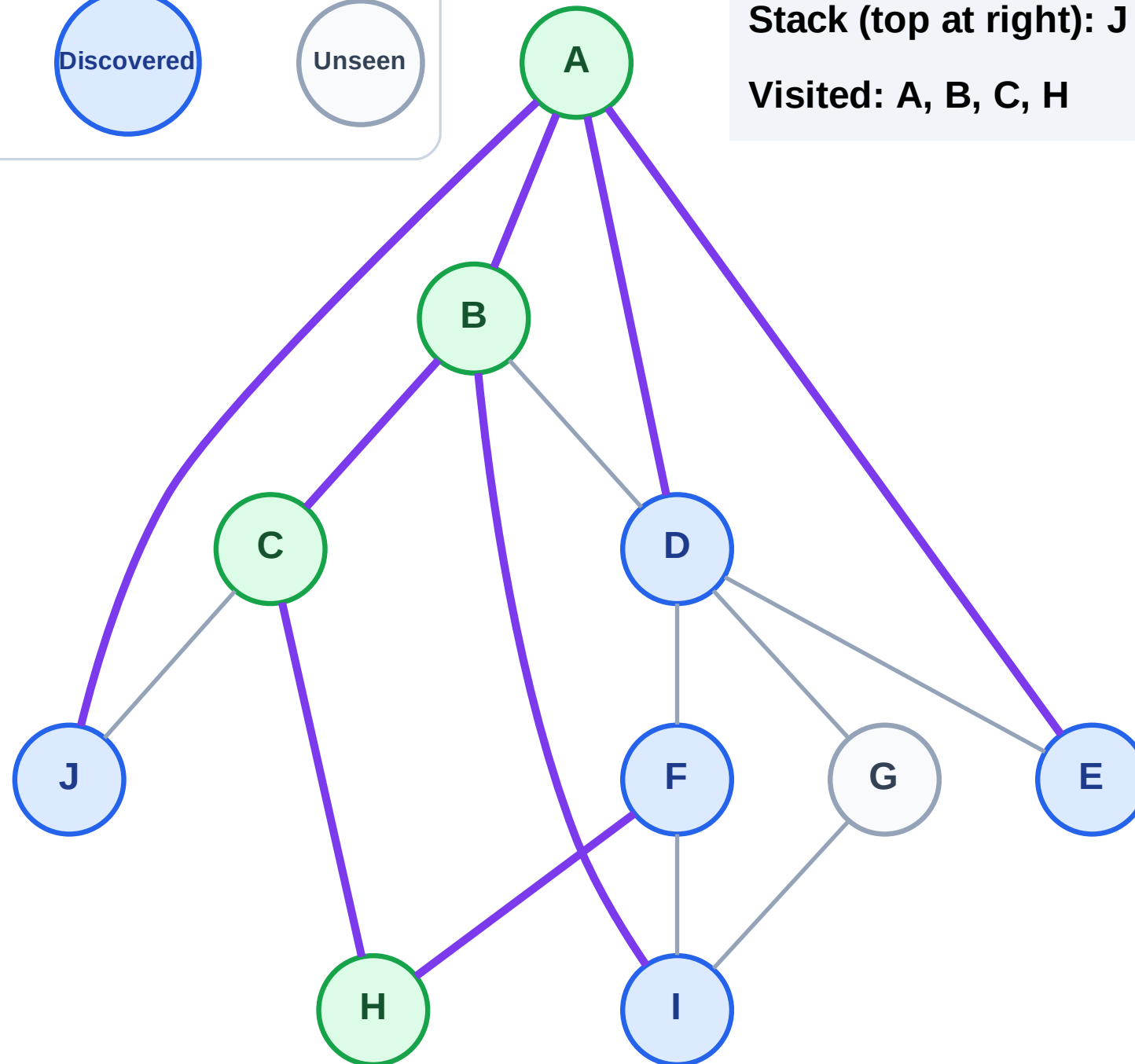
Step 9: Discover F from H

Legend



Stack (top at right): J | E | D | I | F

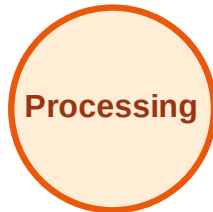
Visited: A, B, C, H



DFS Traversal

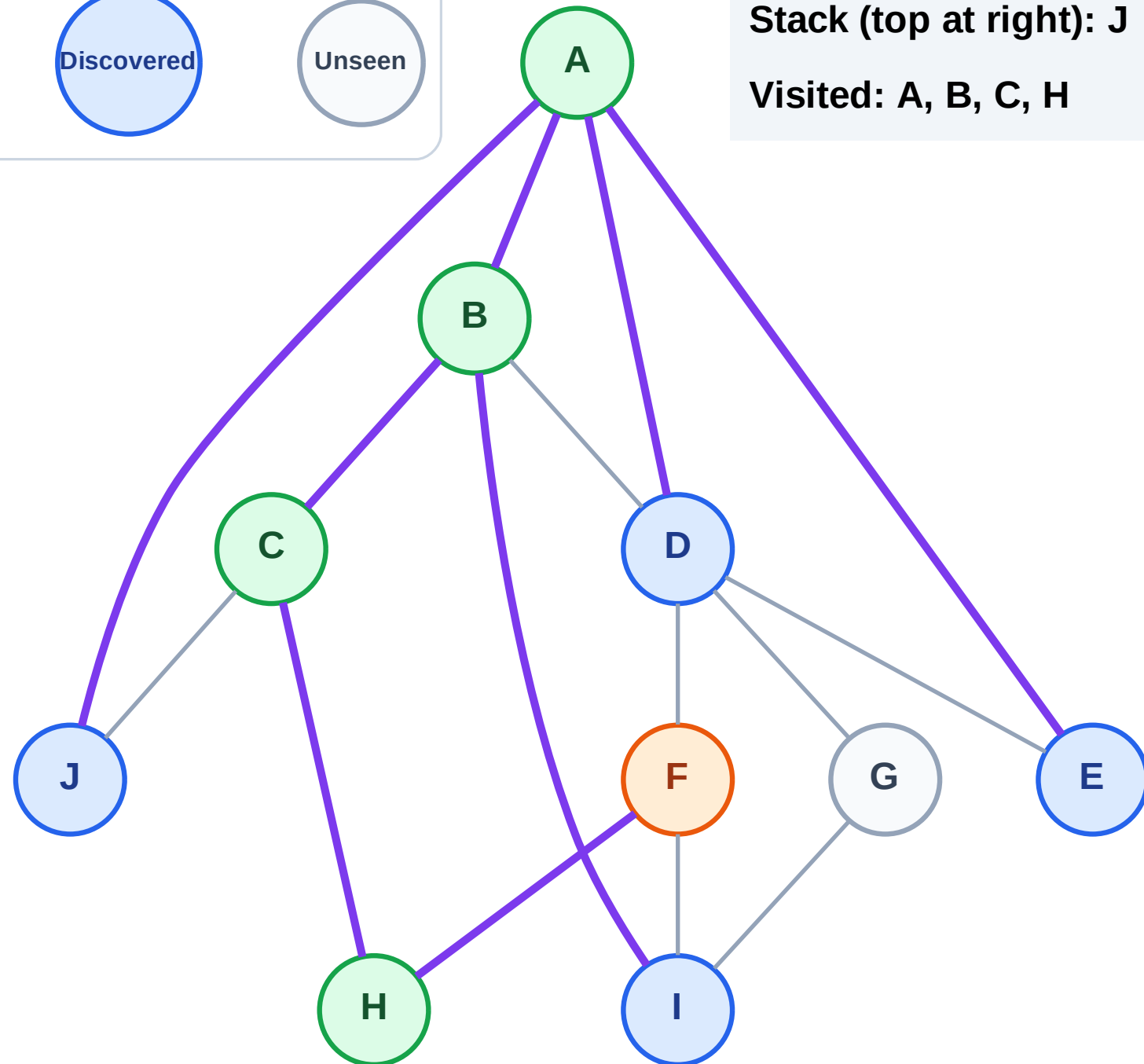
Step 10: Process node F

Legend



Stack (top at right): J | E | D | I

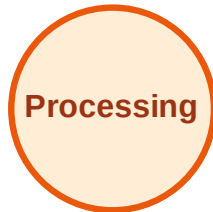
Visited: A, B, C, H



DFS Traversal

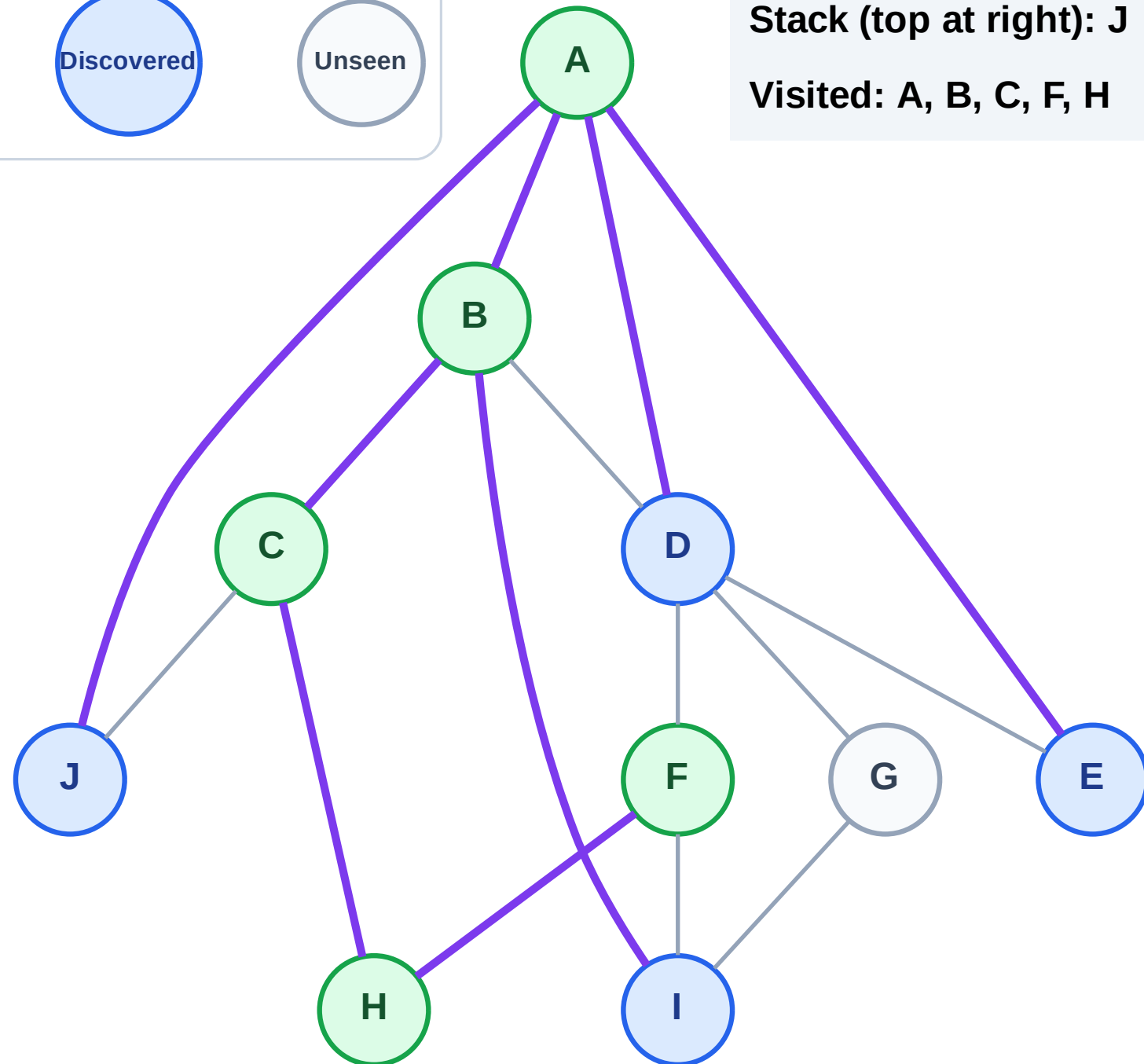
Step 11: No new neighbors from F

Legend



Stack (top at right): J | E | D | I

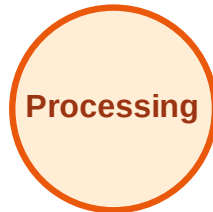
Visited: A, B, C, F, H



DFS Traversal

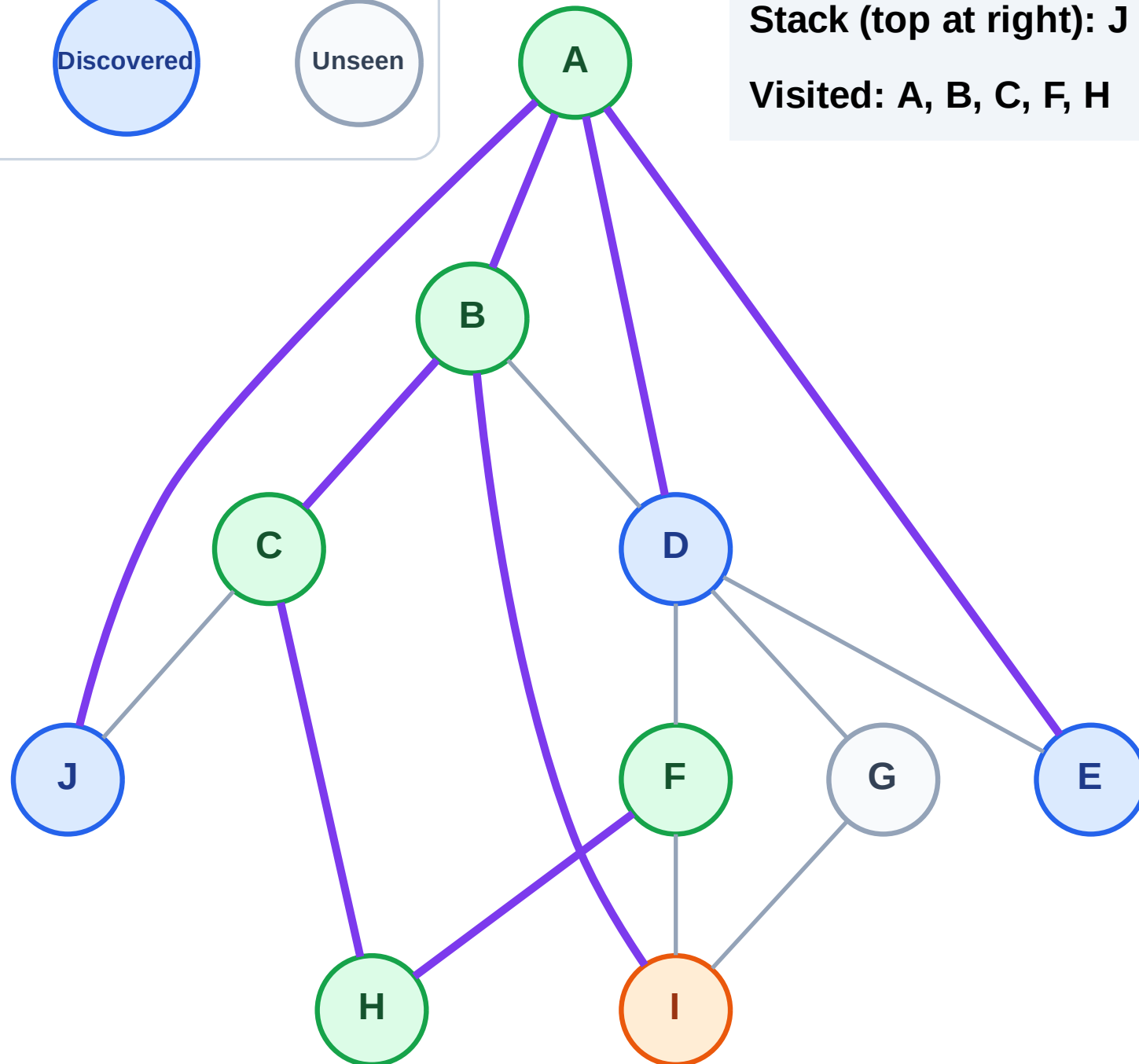
Step 12: Process node I

Legend



Stack (top at right): J | E | D

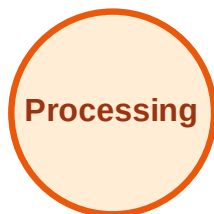
Visited: A, B, C, F, H



DFS Traversal

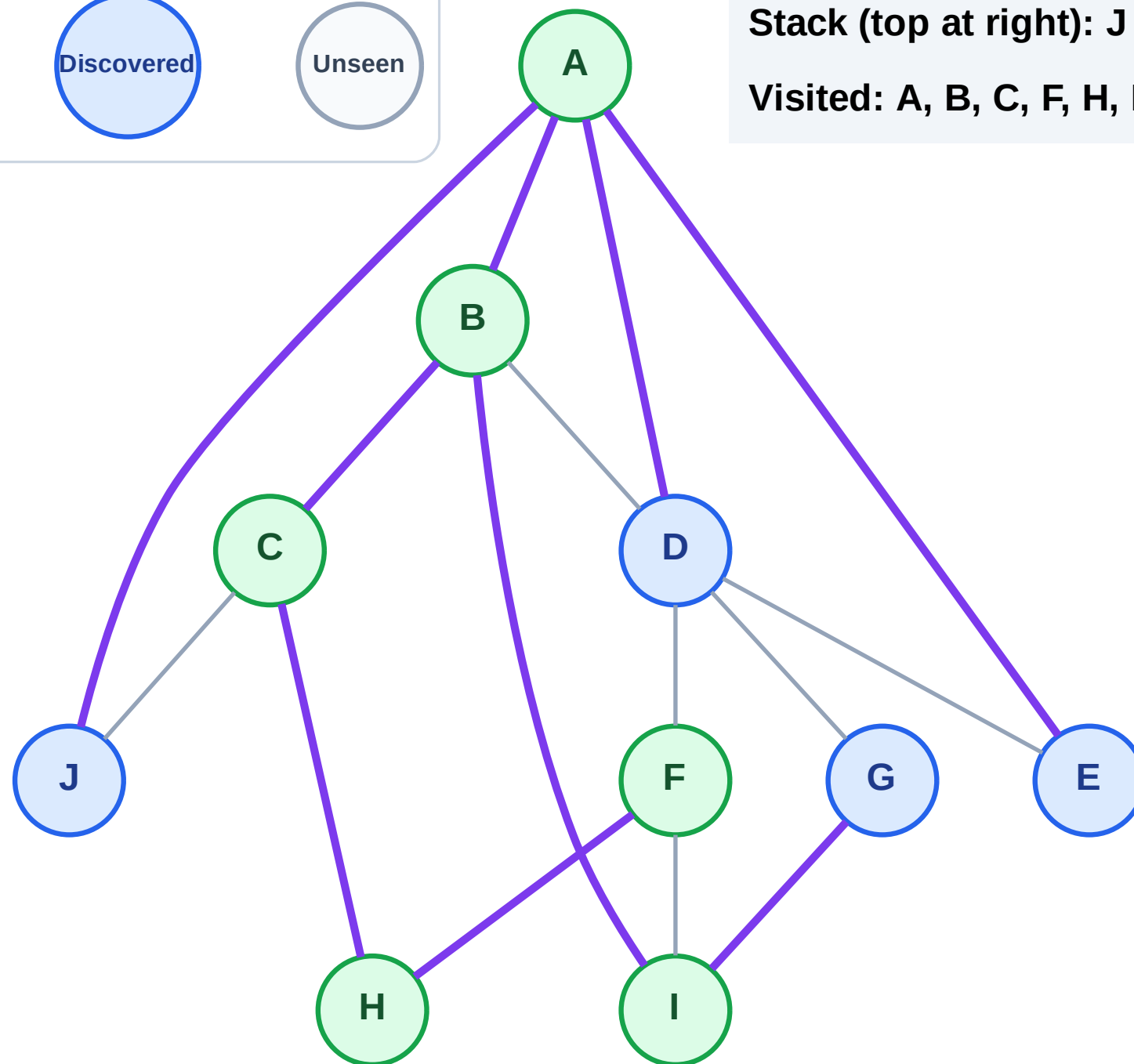
Step 13: Discover G from I

Legend



Stack (top at right): J | E | D | G

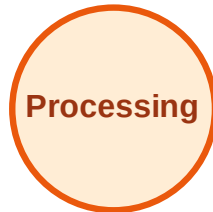
Visited: A, B, C, F, H, I



DFS Traversal

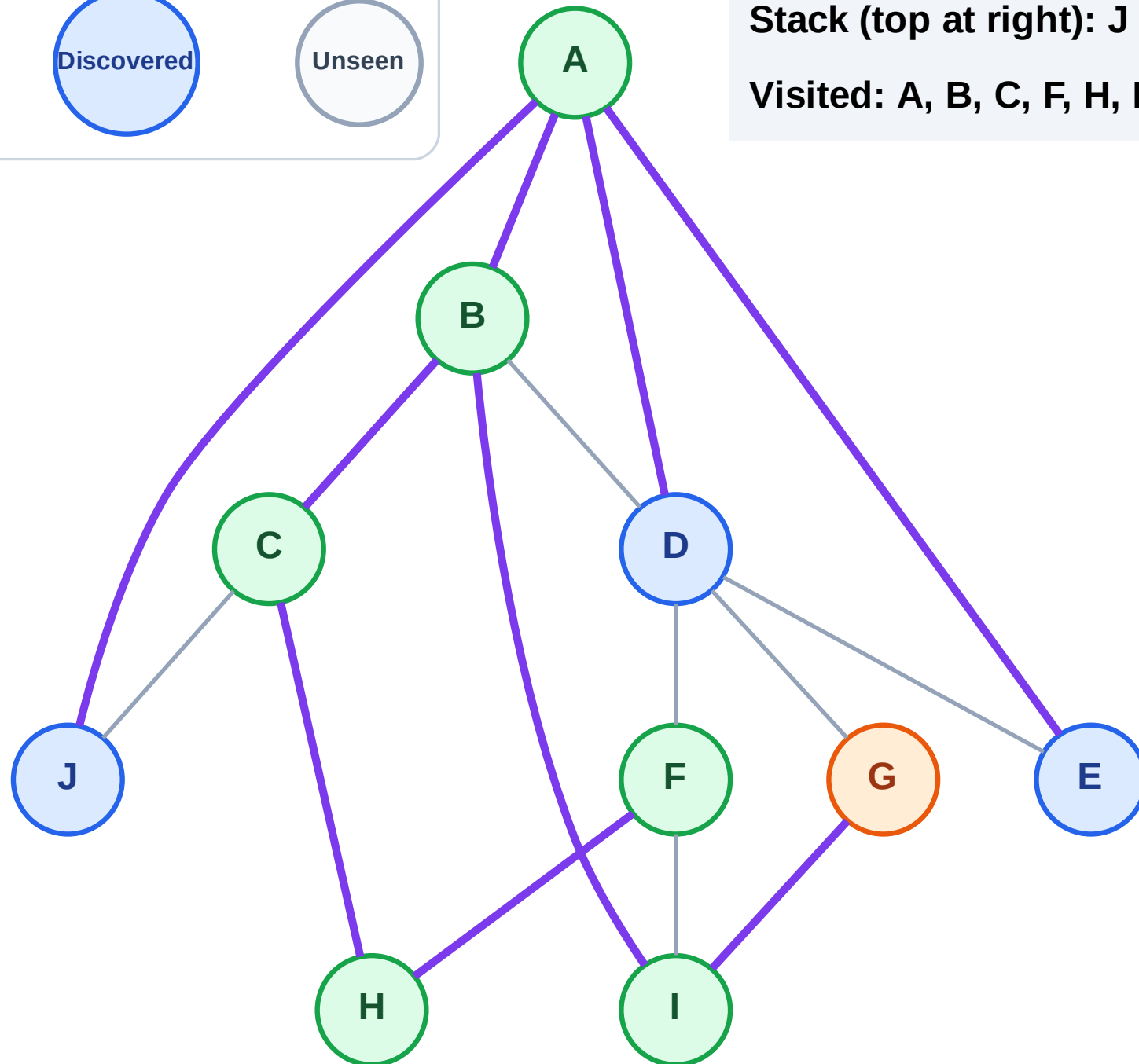
Step 14: Process node G

Legend



Stack (top at right): J | E | D

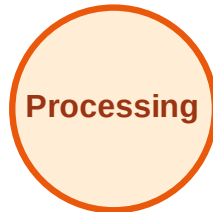
Visited: A, B, C, F, H, I



DFS Traversal

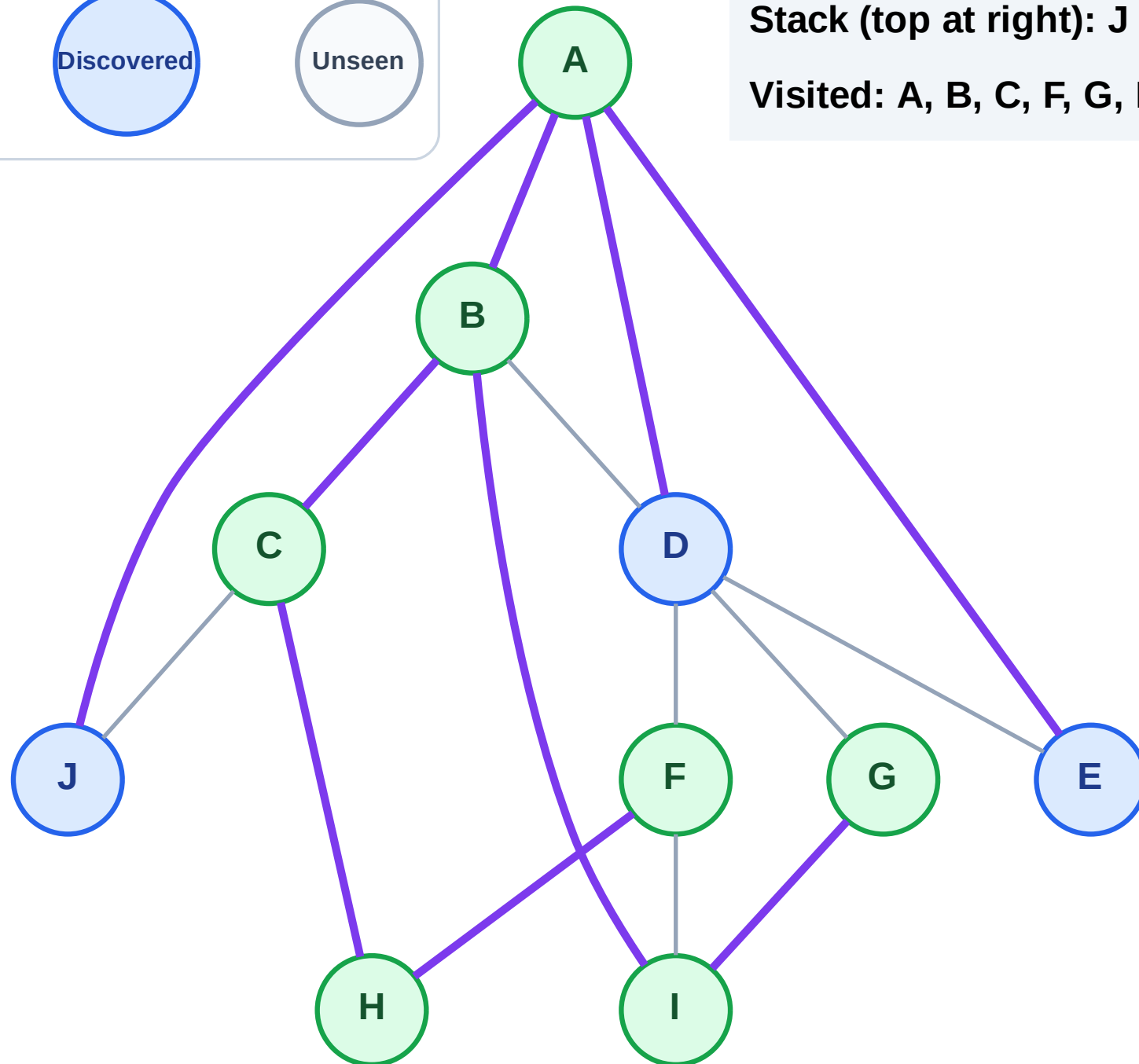
Step 15: No new neighbors from G

Legend



Stack (top at right): J | E | D

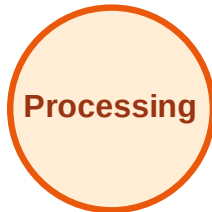
Visited: A, B, C, F, G, H, I



DFS Traversal

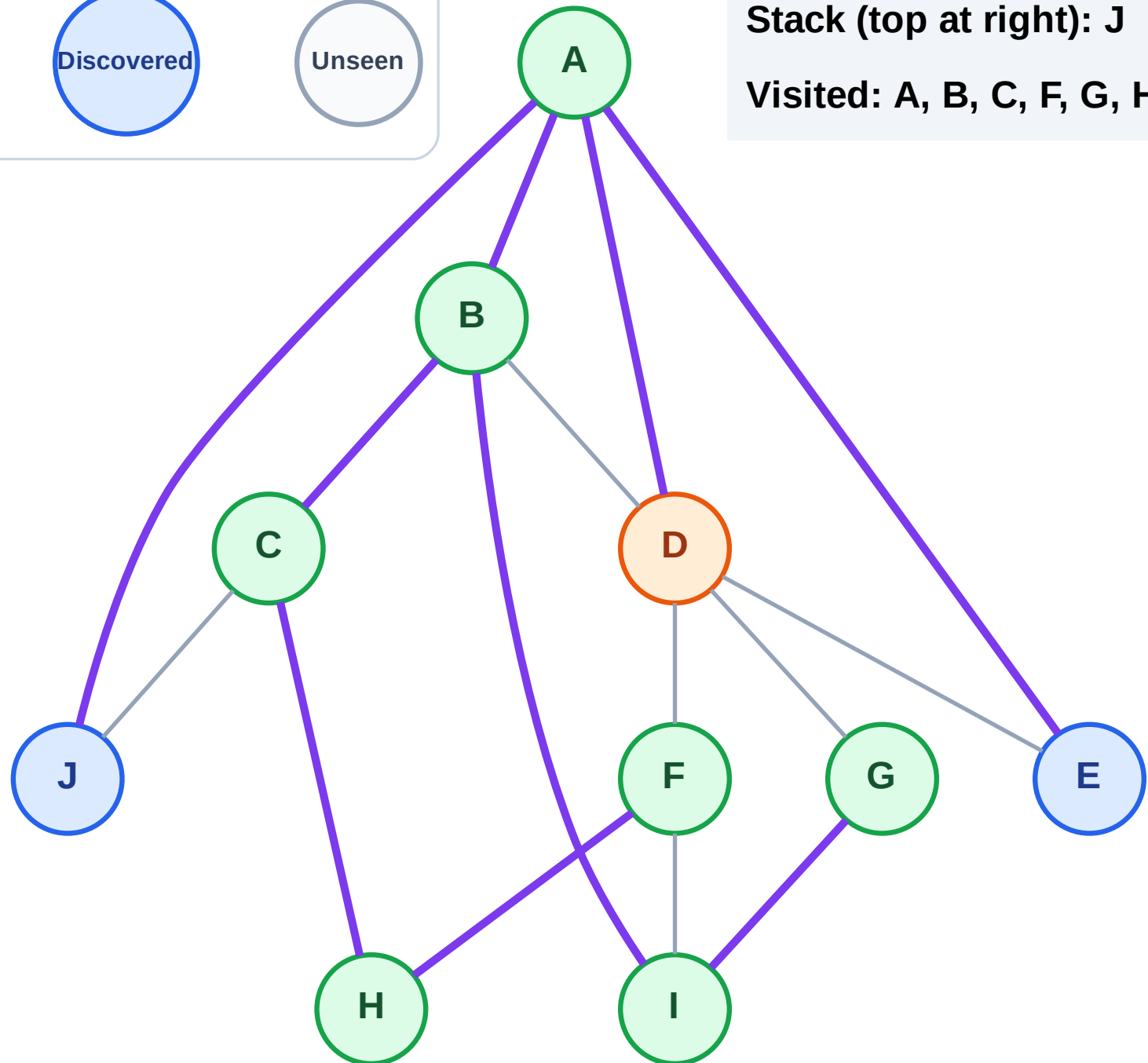
Step 16: Process node D

Legend



Stack (top at right): J | E

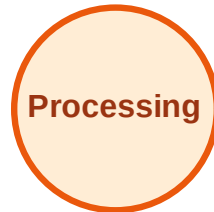
Visited: A, B, C, F, G, H, I



DFS Traversal

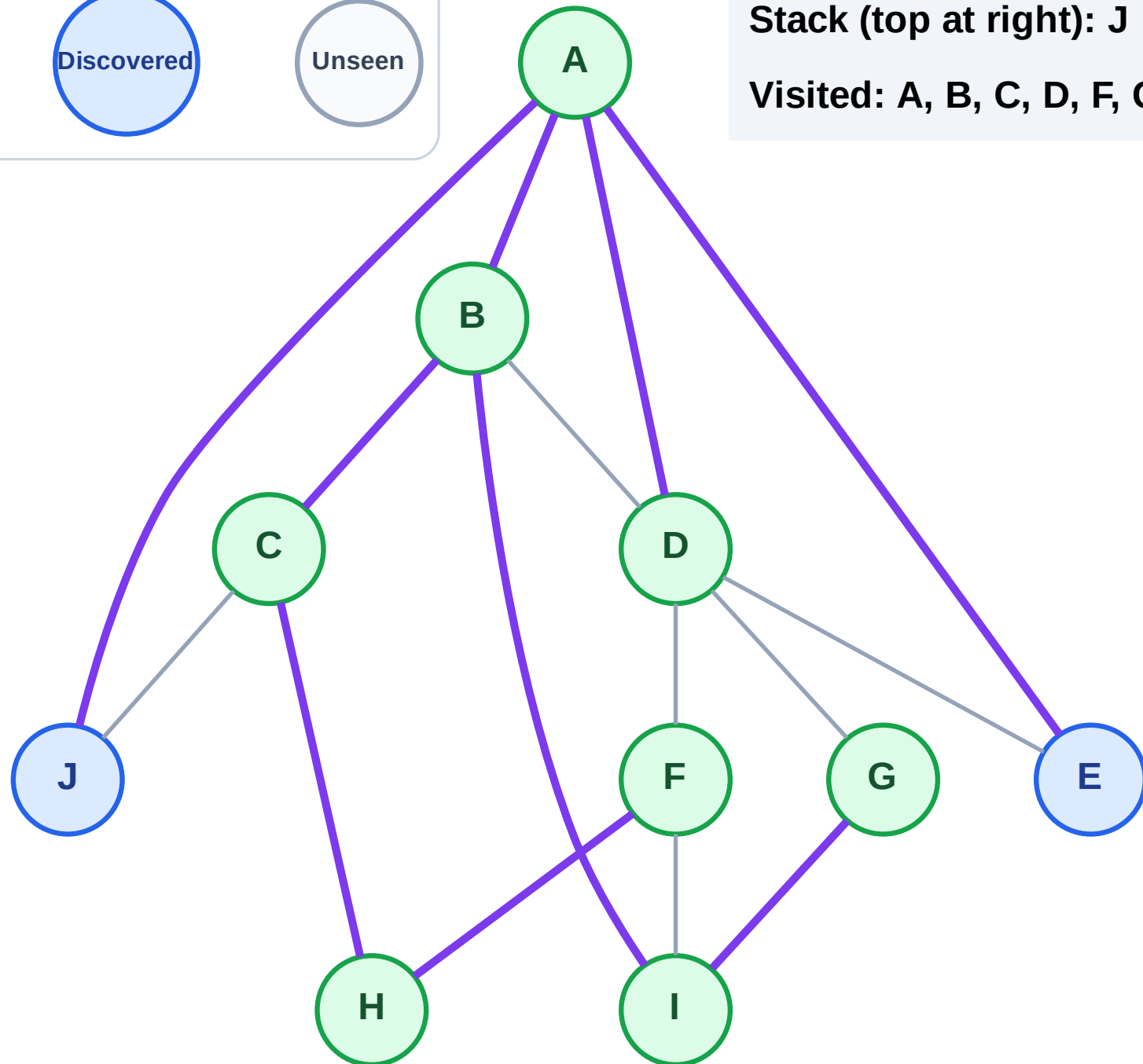
Step 17: No new neighbors from D

Legend



Stack (top at right): J | E

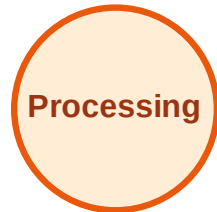
Visited: A, B, C, D, F, G, H, I



DFS Traversal

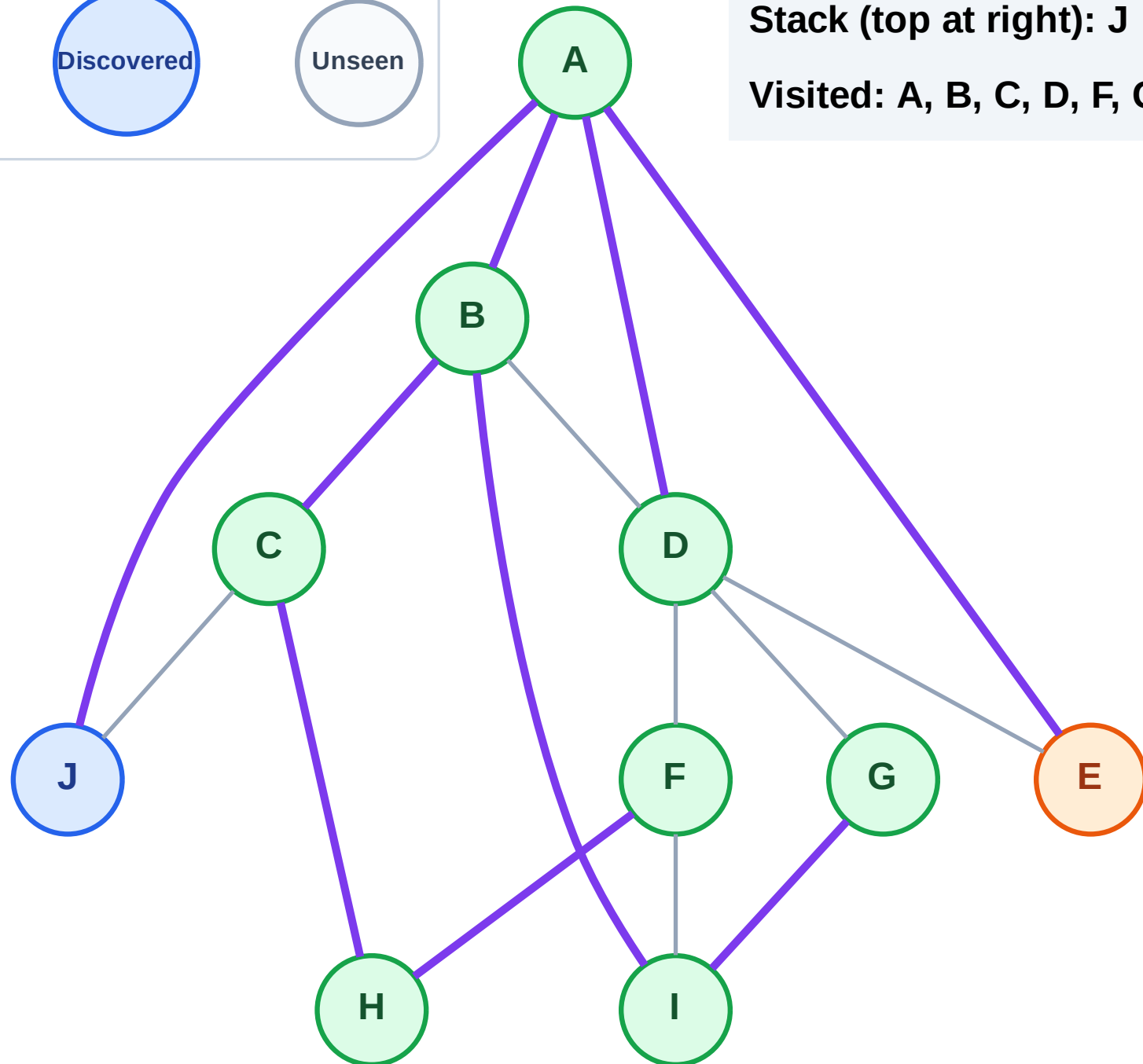
Step 18: Process node E

Legend



Stack (top at right): J

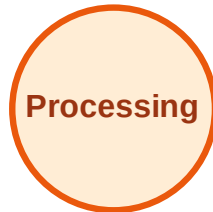
Visited: A, B, C, D, F, G, H, I



DFS Traversal

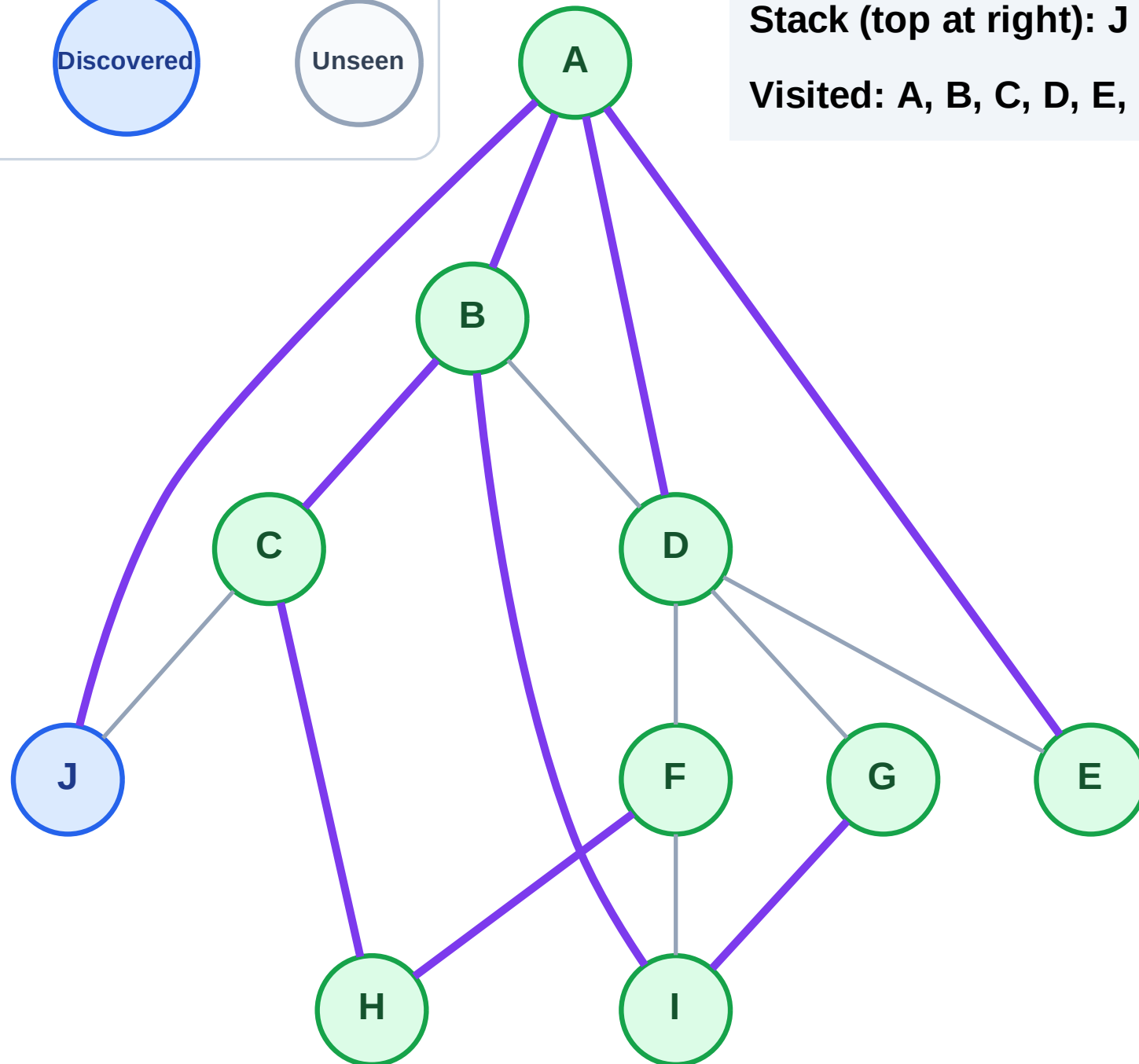
Step 19: No new neighbors from E

Legend



Stack (top at right): J

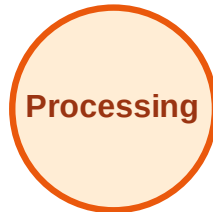
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

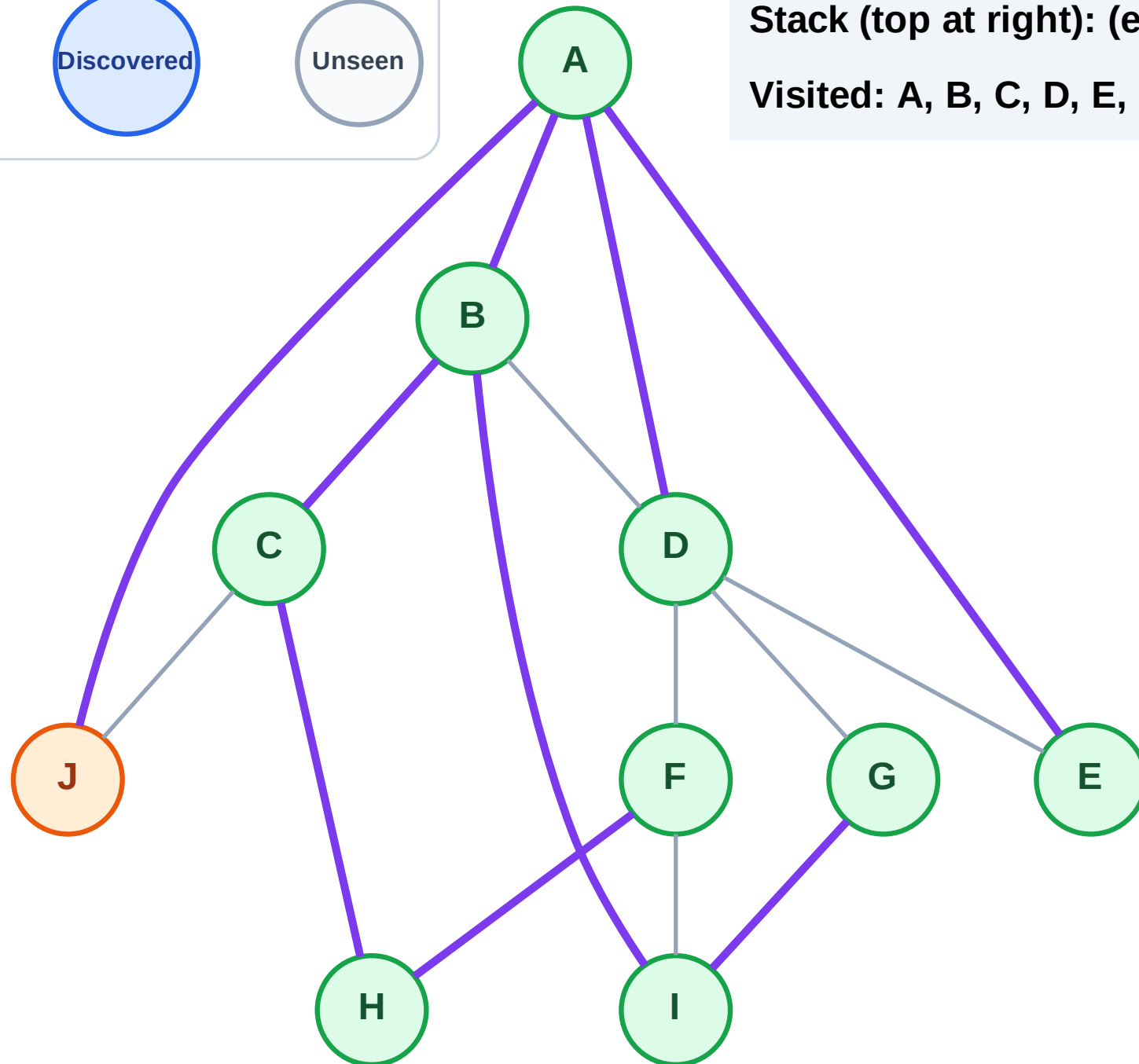
Step 20: Process node J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

